

Building Cloud Infrastructure

Use **VCF Operations** and the **vSphere Client** as a VI administrator to add and configure infrastructure for consumer and management workloads in the VCF private cloud after you deploy or converge to a VCF fleet and a VCF Instance with its management domain.

- You can start adding workload domains to a VCF Instance and individual clusters and hosts to these workload domains.
VCF Operations deploys the vCenter and NSX Manager instances for the added workload domains. You can also enable vSphere Supervisor on the default cluster of the workload domain.
- You can import vCenter instances with their managed ESX hosts as workload domains so that they are included in identity and access management, certificate management, and lifecycle management operations performed centrally in VCF Operations .
- You can also add NSX network services to the management domain and the workload domains of a VCF Instance by deploying NSX Edge clusters, transit gateways, and virtual private clouds (VPCs). Then you can configure workloads to use these network services.
You can place VCF Operations and VCF Automation on NSX overlay or VLAN segments so that they can use the NSX load balancer, or to support planned migration and disaster recovery, or IP mobility.
You can configure consumer workloads to use the network services either in VCF Automation or directly in the vSphere Client.
- If you want to implement vSAN stretched clusters in a workload domain, you can use the SDDC Manager API for automated stretched cluster configuration.

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Getting to Know VCF

To apply this documentation, you must be acquainted with the [VCF Product Overview](#), [VCF Design](#), and [VCF Release Notes](#).

Deploying or Converging to VCF

For information about deploying or converging to a VCF fleet or VCF Instance, see [Deployment, Convergence, and Upgrade](#).

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See [Administration SDKs, APIs, and CLI](#).

Managing ESX Hosts in VMware Cloud Foundation

To add ESX hosts to the global inventory, you must first create a network pool or expand the default network pool created during deployment. Once you have a suitable network pool, you can commission ESX hosts for use in a VCF domain.

During the commissioning process, the ESX hosts are associated with a network pool and added to the global inventory. Newly commissioned ESX hosts appear in the global inventory with a host state of UNASSIGNED. When an ESX host is added to a workload domain, an IP address from the network pool's IP range is assigned to it.

See [VMware Configuration Maximums](#) for information about ESX host maximums.

Note:

VMware Cloud Foundation does not support stateless ESX hosts.

Network Pool Management

When you create a workload domain or add an ESX host or vSphere cluster to an existing VCF domain, you do not need to enter IP addresses manually. Network pools automatically assign static IP addresses to vSAN, NFS, iSCSI, and vMotion VMkernel ports.

A network pool is a collection of subnets within a layer-2 network domain. Depending on the storage option, it includes information about subnets reserved for the vMotion and vSAN, vSAN Storage Client, NFS, or iSCSI networks that are required for adding a host to the global inventory.

Table 821: Information Required for a Network Pool

Storage Being Used	Required Networks in Network Pool
vSAN	vMotion and vSAN
vSAN compute	vMotion and vSAN
vSAN storage (formerly known as vSAN Max)	vMotion, vSAN, vSAN Storage Client
NFS	vMotion and NFS
vSAN and NFS	vMotion, vSAN, and NFS
VMFS on FC	vMotion only or vMotion and NFS
vVols on FC*	vMotion only or vMotion and NFS
vVols on iSCSI*	vMotion and iSCSI
vVols on NFS*	vMotion and NFS

* Starting with VMware Cloud Foundation 9.0 and vSphere Foundation 9.0, the vSphere Virtual Volumes capability, also known as vVols, is deprecated and will be removed in a future release of VMware Cloud Foundation and vSphere Foundation. Support for vSphere Virtual Volumes will continue for critical bug fixes only for versions of vSphere 8.x, VMware Cloud Foundation and vSphere Foundation 5.x, and other supported versions until end-of-support for the respective release.

The network pool contains a range of IP addresses for each network. IP addresses from the ranges are assigned to the VMkernel ports on the ESX host. The use of IP ranges allows you to limit the IP addresses that are consumed from a given subnet. You can add more IP ranges to expand the use of the provided subnet.

A default network pool is created when you deploy VMware Cloud Foundation. This network pool is automatically associated with the management domain.

To create a workload domain with hosts in a different layer-2 network domain than the management domain, you must create a new network pool. Also, if you want to use external NFS or VMFS on FC storage, you must create a new network pool. A network pool can contain both vSAN and NFS networks.

When you create a workload domain with multiple vSphere clusters, those clusters can share a network pool or each cluster can have its own network pool.

Size a Network Pool

Properly sizing a network pool is critical to prevent future issues in the environment due to insufficient IP addresses. Care must be taken when defining the subnets for a network pool as the subnet cannot be changed after it is deployed. The scope of IP addresses used from the defined subnet can be limited by the definition of one or more inclusion ranges. Thus, it is recommended that you begin with defining a larger subnet than what is initially required and utilize the inclusion ranges to limit use. This will provide you the capability to grow with demand as needed.

You begin sizing a network pool by determining the number of ESX hosts that you will have in each vSphere cluster. The minimum number of ESX hosts varies based on the storage type and deployment model. A vSphere cluster can be expanded to the maximum number of ESX hosts supported by vCenter.

Allocate a minimum of one IP address per ESX host plus enough additional IP addresses to account for growth and expansion of the environment. Ensure that the subnet defined provides enough unused IP addresses and that appropriate inclusion ranges are defined. Note that some of the IP addresses within the subnet will be used for other purposes, such as defining the gateway address, firewalls, or other entities. Use care not to conflict with these addresses.

Here are some important considerations for determining the size of your network pool:

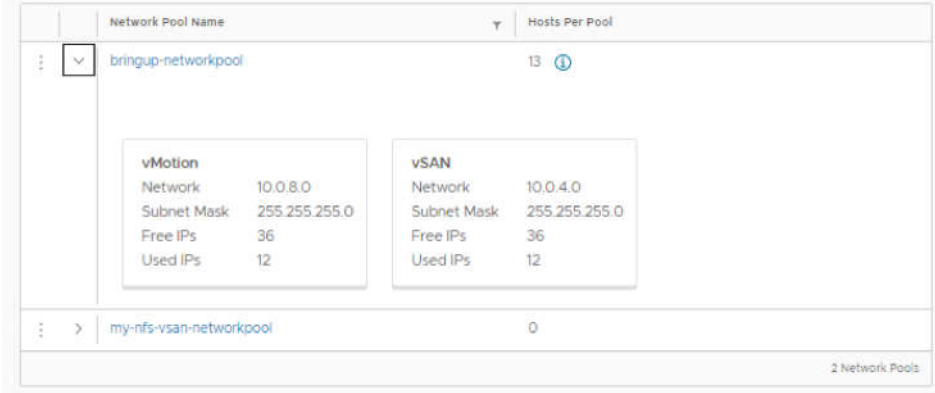
- Type of network architecture.
- Where the default gateway is created.
- Number of hosts that can be placed in each rack or layer-2 network domain.
- Number of hosts required in a vSphere cluster.
- Number of workload domains and cluster you plan on creating.

View Network Pool Details

You can view the network details for a network pool as well as the total number of used and available IP addresses.

To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See [Configuring VCF Single Sign-On](#) and [Linking vCenter instances in VCF Operations](#). As an alternative, you can perform this task using the SDDC Manager UI (<https://<sddc-manager-fqdn>>).

1. In the vSphere Client, click **Global Inventory Lists > Hosts > Network Pools**.
2. Click the arrow to the left of the pool name.



A summary view of the network pools details is displayed.

3. Click the name of a network pool.
A detailed view of the network pool details is displayed, including network types and included IP ranges.

Create a Network Pool

A network pool must include the vMotion network information. Depending on the type of storage you are using, you must provide network information for vSAN, NFS, and iSCSI.

The subnet in a network pool cannot overlap the subnet of another pool.

To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See [Configuring VCF Single Sign-On](#) and [Linking vCenter instances in VCF Operations](#). As an alternative, you can perform this task using the SDDC Manager UI (<https://<sddc-manager-fqdn>>).

Important:

Starting with VMware Cloud Foundation 9.0 and vSphere Foundation 9.0, the vSphere Virtual Volumes capability, also known as vVols, is deprecated and will be removed in a future release of VMware Cloud Foundation and vSphere Foundation. Support for vSphere Virtual Volumes will continue for critical bug fixes only for versions of vSphere 8.x, VMware Cloud Foundation and vSphere Foundation 5.x, and other supported versions until end-of-support for the respective release.

1. In the vSphere Client, click **Global Inventory Lists > Hosts > Network Pools**.
2. Click **Create Network Pool**.
3. Enter a name for the network pool.
4. Select the network type(s).
 - For vSAN ESA/OSA and NFS, you can include both vSAN and NFS network information in the same network pool or create separate network pools for vSAN and NFS.
 - For vSAN storage, select vSAN, vSAN Storage Client, and vMotion.
 - For vSAN compute, select vSAN and vMotion.
 - For VMFS on FC, select vMotion only or vMotion and NFS.
 - For vVols on FC, select vMotion only or vMotion and NFS.
 - For vVols on iSCSI, select vMotion and iSCSI.
 - For vVols on NFS, select vMotion and NFS.
5. Provide the following network information for the selected network type(s).

Option	Description
VLAN ID	Enter a VLAN ID between 1 and 4094.
MTU	Enter an MTU between 1500 and 9000.
Network	Enter a subnet IP address.
Subnet Mask	Enter the subnet mask. Note: You cannot change the subnet after the network pool is created.
Default Gateway	Enter the default gateway.
Included IP Ranges	Enter an IP address range from which an IP address can be assigned to hosts that are associated with this network pool. The IP address range must be from within the specified subnet. You cannot include the IP address of the default gateway in the IP address range. You can enter multiple IP address ranges. Note: Ensure that you have entered the correct IP address range. IP ranges cannot be edited after the network pool is created.

6. Click **Save**.

Add a Network Pool IP Address Range

You can edit a network pool to add a new IP address range.

To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See [Configuring VCF Single Sign-On](#) and [Linking vCenter instances in VCF Operations](#). As an alternative, you can perform this task using the SDDC Manager UI (<https://sddc-manager-fqdn>).

1. In the vSphere Client, click **Global Inventory Lists > Hosts > Network Pools**.
2. Click the vertical ellipsis (three dots) in the network pool row you want to edit and then click **Add additional Network Pool IP Range**.
3. Enter the start and end IP addresses and click **Add**.

Rename a Network Pool

You can rename an existing network pool.

To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See [Configuring VCF Single Sign-On](#) and [Linking vCenter instances in VCF Operations](#). As an alternative, you can perform this task using the SDDC Manager UI (<https://sddc-manager-fqdn>).

1. In the vSphere Client, click **Global Inventory Lists > Hosts > Network Pools**.
2. Click the vertical ellipsis (three dots) in the network pool row you want to edit and then click **Rename Network Pool**.
3. Enter a new name for the network pool.
4. Click **Rename**.

Delete a Network Pool

You can delete a network pool if none of the ESX hosts with an IP address from the pool belong to a workload domain. The default pool created during deployment cannot be deleted.

Ensure that the ESX hosts in the network pool are not assigned to a workload domain. To verify this, view the network pool details and confirm that the **Used IPs** for the network pool is 0.

To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See [Configuring VCF Single Sign-On](#) and [Linking vCenter instances in VCF Operations](#). As an alternative, you can perform this task using the SDDC Manager UI (<https://sddc-manager-fqdn>).

1. In the vSphere Client, click **Global Inventory Lists > Hosts > Network Pools**.
2. Click the vertical ellipsis (three dots) in the network pool row you want to edit and then click **Delete**.
3. Click **Delete**.

View ESX Host Inventory

The Hosts page in the vSphere Client displays details about all the ESX hosts in the global inventory, including CPU utilization and memory usage across all hosts, as well as the total number of hosts used and unallocated.

For each ESX host, the Hosts page displays details, including name, status, vSphere cluster, and CPU and memory usage.

The Hosts page also provides controls for commissioning hosts.

1. In the vSphere Client, click **Global Inventory Lists > Hosts**.
2. Click **Assigned Hosts** to see a list of ESX hosts that are assigned to a workload domain.

You can view details about each ESX host, including name, vSphere cluster, and CPU and memory usage. Click the host name to view more details about the host in the vSphere inventory. Click the vSphere cluster name to view more details about the cluster in the vSphere inventory.

- Click **Unassigned Hosts** to see a list of commissioned ESX hosts that are available to create a new workload domain or add to an existing workload domain.

You can view details about commissioned ESX hosts and also commission and decommission ESX hosts from the **Unassigned Hosts** tab.

Commission ESX Hosts

Adding ESX hosts to the global inventory is called commissioning. You must commission ESX hosts before you can use them to create a new workload domain or add them to an existing VCF domain.

Prepare your hosts. See [Preparing ESX Hosts for VMware Cloud Foundation or vSphere Foundation](#).

Ensure that each ESX host you commission meets the following criteria:

- Hosts for vSAN-based workload domains are vSAN-compliant and certified on the VMware Hardware Compatibility Guide.
 - Hosts for NFS-based workload domains are certified on the VMware Hardware Compatibility Guide.
 - Hosts for VMFS on FC-based workload domains are certified on the VMware Compatibility Guide. In addition, the hosts must have supported FC cards (Host Bus Adapters) and drivers installed and configured. For compatible FC cards, see the VMware Compatibility Guide.
 - Important:** The VMFS datastore must be mounted on all the hosts before commissioning.
 - Hosts for vVols-based workload domains are certified on the VMware Hardware Compatibility Guide.
 - For vVols on FC-based workload domains, ensure that all ESX hosts have access to the FC array before launching the workflow.
 - For vVols on NFS-based workload domains, ensure that all ESX hosts must be able to reach the NFS server from the NFS network assigned in the IP pool.
 - For vVols on iSCSI-based workload domains, ensure that the iSCSI software initiator must be enabled on each ESXi host and the VASA provider URL must be listed as the dynamic target.
- Important:**

Starting with VMware Cloud Foundation 9.0 and vSphere Foundation 9.0, the vSphere Virtual Volumes capability, also known as vVols, is deprecated and will be removed in a future release of VMware Cloud Foundation and vSphere Foundation. Support for vSphere Virtual Volumes will continue for critical bug fixes only for versions of vSphere 8.x, VMware Cloud Foundation and vSphere Foundation 5.x, and other supported versions until end-of-support for the respective release.

- Each host must have the correct number of physical NICs for the desired network configuration for the host. Hosts can have a single physical NIC or multiple physical NICs. For hosts with multiple physical NICs, only one can be connected to a standard switch, and the NIC must support all management traffic.
- All physical NICs must have a minimum speed of 10 Gbps.
- Management IP address is configured on a VMkernel interface connected to the Management Network on the standard switch.
- Host has the drivers and firmware versions specified in the VMware Hardware Compatibility Guide.
- A supported version of ESX is installed on the host. See the *VMware Cloud Foundation Release Notes* for information about supported versions.
- Has both forward and reverse lookup entries (FQDN) configured on the DNS server.
- Host name must be same as the FQDN.
- Self-signed certificate regenerated based on FQDN of host. See [Regenerate the Self-Signed Certificate on ESX Hosts](#).
- Hardware health status is healthy without any errors.
- All disk partitions on HDD and SSD are deleted.
- A network pool must be created and available before host commissioning.
- The network pool must support the storage type you select for a host (vSAN, vSAN ESA, vSAN Storage Client, NFS, VMFS on FC, vVols). See [Network Pool Management](#).
- Host is configured with an appropriate gateway. The gateway must be part of the management subnet.

To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See [Configuring VCF Single Sign-On](#) and [Linking vCenter instances in VCF Operations](#). As an alternative, you can perform this task using the SDDC Manager UI (<https://<sddc-manager-fqdn>>).

You can add ESX hosts individually or use a JSON template to commission multiple hosts at once. You can also run multiple commission hosts tasks at the same time.

After you specify host details and select the network pool to associate the hosts with, the platform validates and commissions each host. Each host is added to the free pool and is available for VCF domains and vSphere clusters.

- Hosts that use vSAN storage can only be used with vSAN-based clusters.
 - Hosts that are commissioned for vSAN HCI with ESA enabled can only be used for vSAN ESA clusters.
 - Hosts that are commissioned for vSAN HCI without ESA enabled can only be used for vSAN OSA clusters.
 - Hosts that are commissioned for vSAN Storage (formerly vSAN Max) can only be used for vSAN storage clusters.
 - Hosts that are commissioned for vSAN Compute Cluster can only be used for vSAN compute clusters.
- Hosts that use NFS storage can only be used with NFS-based clusters.
- Hosts that use VMFS on FC storage can only be used with VMFS on FC-based clusters.
- Hosts that use vVols storage can only be used with vVols-based clusters.

See [VMware Configuration Maximums](#) for information about the maximum number of hosts you can commission at one time and the maximum number of commission hosts tasks that you can run in parallel.

If you used ESX hosts with certificates generated by an external CA for deployment, hosts used for expanding VCF domains or for creating workload domains must also use certificates generated by an external CA. See [Configure ESX Hosts with Signed Certificates](#).

Note:

When you add a host during host commissioning, the platform temporarily activates SSH on the host in order to retrieve the host key. After it retrieves the host key, the platform deactivates SSH on the host.

With VMware Cloud Foundation 9.0.1 and later, you can use non-certified ESA disks during vSAN ESA cluster configuration for proof-of-concept purposes only. For more information, see [KB 408300](#).

- In the vSphere Client, click **Global Inventory Lists > Hosts > Unassigned Hosts**.
- Click **Commission Hosts**.
- On the **Checklist** page, review the prerequisites and confirm by clicking **Select All**.
- Click **Proceed**.
- Select whether you want to add hosts one at a time or import multiple hosts at once from a JSON file.

Option	Description
Add new	Enter the following information for the host you want to add: - Host FQDN The host FQDN must match the host name (including capitalization) provided during DNS configuration. - Storage Type (vSAN, NFS, VMFS on FC, or vVol) for principal storage - If you select vSAN as the storage type: - Select the vSAN Type (vSAN HCI, vSAN Storage, or vSAN Compute Cluster). - For vSAN HCI choose whether or not to use the vSAN Express Storage Architecture (ESA). - If you select vVols as the storage type, specify the vVols storage protocol type (FC, iSCSI, or NFS). - Network Pool Name (select a network pool from the drop-down menu) - User Name and Password (root credentials for the ESX host)

Option	Description
	Click Add . You can now add more ESX hosts or proceed to the next step.
Import	- To download the JSON template, click the link. - Open the JSON template file and enter information about the hosts to add. - Host FQDN - Username and password (ESX root credentials) - Storage type (VSAN, VSAN_REMOTE, VSAN_ESA, VSAN_MAX, NFS, VMFS_FC, VVOL) for principal storage Note: Use VSAN_REMOTE for a vSAN compute cluster. Use VSAN_MAX for a vSAN storage cluster. - Network pool name Note: You only need to provide the vVols storage protocol type (FC, ISCSI, or NFS) when you are using VVOLS as the storage type - To locate and select the JSON file containing host information, click Browse. - Click Upload.

The ESX host or hosts appear in the **Hosts Added** section.

6. Verify that the server fingerprint is correct for each host and then activate the **Confirm All Finger Prints** toggle.

7. Click **Validate All**.

The platform validates the host information you provided. Each host is marked as **Valid** or **Invalid**.

For invalid hosts, you can correct the problem and validate again, or select the host and click **Remove** to proceed with commissioning the valid hosts.

8. Click **Next**.

9. Activate or deactivate **Skip failed hosts during commissioning**.

This setting is enabled by default. When enabled, hosts that fail are skipped and commissioning continues on the remaining hosts.

10. Click **Commission**.

The Hosts page appears, and the status of the commission task is displayed in the Tasks window.

Note: Multiple VMkernels are created to test the vMotion network. If MAC address filtering is enabled on your physical infrastructure, this may cause issues such as vMotion network connectivity validation failure.

The commissioned hosts are added to the Unassigned Hosts table.

Decommission ESX Hosts

Removing ESX hosts from the global inventory is called decommissioning. If you want to re-use a host in a different VCF domain, you must decommission, re-image, and commission the host before adding it to the VCF domain.

The hosts that you want to decommission must not be assigned to a VCF domain. If a host is assigned to a VCF domain, you must remove it before you can decommission it.

To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See [Configuring VCF Single Sign-On](#) and [Linking vCenter instances in VCF Operations](#). As an alternative, you can perform this task using the SDDC Manager UI (<https://sddc-manager-fqdn>).

See [VMware Configuration Maximums](#) for information about the maximum number of hosts you can decommission at one time and the maximum number of decommission hosts tasks that you can run in parallel.

1. In the vSphere Client, click **Global Inventory Lists > Hosts > Unassigned Hosts**.

2. Select one or more hosts and click **Decommission Selected Hosts**.

3. Activate or deactivate **Skip failed hosts during decommissioning**.

This setting is enabled by default. When enabled, hosts that fail are skipped and decommissioning continues on the remaining hosts.

4. Click **Confirm**.

If the decommissioned host was previously part of a VCF domain, you must re-image it before commissioning it again and adding it to a VCF domain. If the decommissioned host was not part of a VCF domain, you do not have to re-image it before commissioning it again.

Managing VCF Domains in VMware Cloud Foundation

A VCF domain represents a logical unit of application-ready infrastructure that groups ESX hosts managed by a vCenter instance with specific characteristics according to VMware recommended practices. VCF supports two types of VCF domains; the management domain and workload domains.

The management domain is created during the deployment or convergence process by VCF Installer and contains the management components for a VCF instance, including SDDC Manager, vCenter, NSX Manager, and ESX hosts. After the management domain is created, you can deploy or import workload domains to run customer workloads. Workload domains include a vCenter and NSX Manager (which can be shared with other workload domains), as well as one or more vSphere clusters with ESX hosts.

For more information about VCF domains, see the [VMware Cloud Foundation Overview](#) and [Design Guide](#).

After creating or importing a workload domain in VCF Operations, you can add or remove vSphere clusters and ESX hosts using the vSphere Client to support the resource and availability needs of your organization.

Create a New Workload Domain

Creating a new workload domain adds a logical pool of compute, network, and storage infrastructure to a VCF instance.

General requirements:

- A vSphere Lifecycle Manager cluster image must be available for the default vSphere cluster of the workload domain. See [Managing vSphere Lifecycle Manager Images for VMware Cloud Foundation](#).
- If the hosts in the management domain have an express patch applied, the hosts for the new workload domain must be imaged with the same express patch.
- Hosts must be commissioned with the target principal storage type. See [Managing ESX Hosts in VMware Cloud Foundation](#).
- If you are using vVols principal storage type, add a VASA provider to the inventory. See [Add a VASA Provider for vVols Storage in Workload Domains](#).

Important:

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- If you are using vSAN principal storage type, SSD disks or NVMe disks without pre-existing disk partitions must be available with each host.

Networking requirements:

- A static IP pool or a DHCP server configured and advertising IP addresses on the workload domain's NSX host overlay (HOST TEP) VLAN.

- Each host must have at least one physical NIC, with a minimum of 10 Gbps speed, connected to a standard switch.
- A network pool with free IP addresses must be available.
- The IP addresses and Fully Qualified Domain Names (FQDNs) for the vCenter and NSX Manager instances must be resolvable over DNS

For information about importing an existing vCenter environment as a workload domain, see [Import an Existing vCenter to Create a Workload Domain](#).

VMware Cloud Foundation supports creating workload domains with a vSphere Distributed Switch with link aggregation control protocol (LACP) enabled, but you must use the [SDDC Manager API](#) to do so.

1. Log in to VCF Operations.
2. In VCF Operations, select **Inventory > Detailed View**.
3. Expand **VCF Instances** and browse to the VCF instance in which you want to create a new workload domain.
4. Click **Add Workload Domain > Create New**.
5. Review the prerequisites, click **Select All**, and click **Proceed**.
6. Enter the General Information details and click **Next**.

Option	Description
Workload Domain Name	Type a name for the workload domain, such as <code>sfo01</code> . The name must contain between 3 and 20 characters. It is good practice to include location information in the name since resource object names (such as vSphere Distributed Switch names) are generated based on the workload domain name.
Enable vSphere Supervisor	This setting is activated by default, but can be deactivated. When activated, the workload domain is enabled for vSphere Supervisor with NSX VPC Networking, creating a Supervisor, a single Supervisor control plane VM, and a single vSphere zone. After the workload domain is complete, you must deploy an NSX Edge cluster with an active-standby tier-0 gateway to complete Supervisor activation, unless you are using an existing NSX Manager instance with a compatible NSX Edge cluster. See Configure Centralized Network Connectivity with Edge Clusters for information about deploying an NSX Edge cluster. Use the vSphere Client to scale out, upgrade, monitor, and configure vSphere Supervisor after deployment. Click Learn More and review the vSphere Supervisor documentation for more information.
SSO Domain Name	Enter the vCenter Single Sign-On (SSO) domain name, for example, <code>example.domain.local</code> . This creates a workload domain that is isolated from the other workload domains in your VCF instance.
Password Creation	Select Auto-generate my passwords for newly installed appliances to generate random complex passwords for any appliances that get deployed during workload domain creation. You can retrieve auto-generated passwords using the SDDC Manager API . If choose not to auto-generate passwords, set and confirm the administration password for the SSO

Option	Description
	administrator account. This is the password for the user <code>administrator@example.domain.local</code> . Requirements: • 8-20 characters. • No spaces. • At least one lowercase character. • At least one uppercase character. • At least one digit. • At least one special character. • No dictionary words.

7. Enter the vCenter details and click **Next**.

Option	Description
vCenter FQDN	Enter the FQDN for the workload domain's vCenter. Note: The vCenter for a workload domain deployed in the management domain.
vCenter Root Password	Enter and confirm a vCenter root password. Requirements: • No dictionary words. • 15-20 characters. • At least one lowercase character. • At least one digit. • No spaces. • At least one special character. • At least one uppercase character. Note: This is not required if you chose to auto-generate passwords.

8. Enter the Cluster details and click **Next**.

Option	Description
Cluster Name	Enter a name for the vSphere Cluster.
vSphere Zone Name	If you enabled vSphere Supervisor, enter a vSphere Zone name. The vSphere Zone is created and the cluster is added to it. Note: This workflow creates a single vSphere Zone. You can use the vSphere Client to create additional zones.

9. Select a cluster image and click **Next**.

Note: If the cluster image contains a different version of a vendor add-on or component than what is installed on the ESX hosts you add to the cluster, the hosts will be remediated to use the cluster image during cluster creation.

10. Enter the NSX Manager details and click **Next**.

Choose whether to create a new NSX Manager instance or join an existing NSX Manager instance. If you already have an NSX Manager instance for a different workload domain, you can reuse that NSX Manager instance or create a new one.

If you are using an existing NSX Manager instance, select it. If you are creating a new NSX Manager instance, enter the following information.

Option	Description
Deployment Size	• Standard: Deploys single-node NSX Manager requiring fewer resources. • High-Availability (recommended): Deploys a three-node NSX Manager cluster to enhance the availability of management and control plane functions, as well as minimize downtime during upgrades.
Appliance Size	• Medium • Large • Extra Large
Appliance 1 FQDN	Enter the FQDN for the first NSX Manager node.
Appliance 2 FQDN	Enter the FQDN for the second NSX Manager node. This is not applicable if you chose Standard deployment size.
Appliance 3 FQDN	Enter the FQDN for the third NSX Manager node. This is not applicable if you chose Standard deployment size.
Appliance Cluster FQDN	Enter the NSX Manager cluster FQDN.
Administrator Password	Enter and confirm an NSX administrator password. Requirements: • At least five different characters. • At least one lowercase character. • At least one digit. • No more than four monotonic or sequential characters. • At least 12 characters. • Cannot be palindrome. • At least one special character. • At least one uppercase character. • No dictionary words. Note: This is not required if you chose to auto-generate passwords.
Auditor Password	Enter and confirm an NSX auditor password. Requirements: • At least five different characters. • At least one lowercase character. • At least one digit. • No more than four monotonic or sequential characters. • At least 12 characters. • Cannot be palindrome. • At least one special character. • At least one uppercase character. • No dictionary words.

Option	Description
	Note: This is not required if you chose to auto-generate passwords.
Configure Network Connectivity and VPC External IP Blocks	Select one of the options: • Centralized Connectivity • Distributed Connectivity Note: vSphere Supervisor requires Centralized Connectivity. If you select Centralized Connectivity , configure it after the workload domain is created to make it VPC-ready. See Configure Centralized Network Connectivity with Edge Clusters . If you select Distributed Connectivity , enter the following: • VLAN ID: A dedicated VLAN network is required for the distributed gateway's external connectivity. • Gateway CIDR IPv4 Address: The gateway CIDR IP address should be the same as the VLAN default gateway configured in the external router (TOR). • External IP Block: Advertised CIDRs that allow outside connectivity to VPC workloads either on public subnets or through external IPs. The external IP block must be the same as, or a sub-block of, the configured Gateway CIDRs. • Private - Transit Gateway IP Block: Private CIDRs that are available for inter-VPC (Virtual Private Cloud) communication and are not advertised by the Transit Gateway to the outside datacenter. After the workload domain with distributed connectivity is created, it is VPC-ready. For more information about VPCs, see Managing Virtual Private Clouds in vCenter .

11. Enter the Storage details and click **Next**.

Option	Description
vSAN	Select to configure a vSAN-based workload domain. Select Enable vSAN ESA to configure a vSAN ESA workload domain. If you do not select this option, the workload domain is configured with vSAN OSA. Note: vSAN storage clusters (formerly known as vSAN Max) require vSAN ESA
NFS	Select to configure an NFS-based workload domain.
VMFS on FC	Select to configure a Fibre Channel-based workload domain
vVol	Select to configure a vVol-based workload domain.

12. Enter the vSAN Storage details and click **Next**.

This screen only displays if you selected vSAN storage. The vSAN storage options are different for vSAN OSA and vSAN ESA.

vSAN OSA:

Option	Description
Failures to Tolerate	Specify the level of availability you want configured for the cluster. Your selection determines the minimum number of required ESX hosts.
Space efficiency	Select the check box to enable vSAN deduplication and compression

vSAN ESA:

Option	Description
vSAN Cluster Type	vSAN HCI: Provides storage and compute resources. vSAN Storage: Provides storage resources only. You can mount a vSAN Storage datastore on other vSAN ESA or vSAN compute clusters.
Network Option (vSAN Storage only)	Select the vSAN Storage Client Network checkbox to isolate client or application network traffic from vSAN Storage internal traffic by utilizing dedicated VMkernel ports backed by physical network ports.
Storage Type	Local vSAN datastore. This setting cannot be modified.
Storage Policy	Auto-policy management. This setting cannot be modified. Based on the type of cluster and number of hosts, vSAN creates and assigns a default datastore policy for best capacity utilization after the cluster configuration is completed. Policy details can be viewed in the vSphere Client (Policies and Profiles > VM Storage Policies).

13. Enter the NFS Storage details and click **Next**.

This screen only displays if you selected NFS storage.

Option	Description
Datastore Name	Enter a name for the NFS datastore. The datastore name must contain 3-80 characters and cannot contain any leading or trailing whitespace. If you using a previously mounted NFS share, the datastore will be automatically renamed to original datastore name.
Folder	Enter the path to the NFS share. The folder location must contain 3-255 characters and the path cannot contain any whitespace.
NFS Server IP Address	Enter the IPv4 address of the NFS server.

14. Enter the FC Storage details and click **Next**.

This screen only displays if you selected FC storage.

Option	Description
Datastore Name	Enter the name of the VMFS on FC datastore.

Option	Description
	Datastore name must contain 3-80 characters and cannot contain any leading or trailing whitespace.

15. Enter the vVol Storage details and click **Next**.

This screen only displays if you selected vVols storage.

Option	Description
VASA Protocol Type	Select a VASA protocol type. vVols supports FC, NFS, and iSCSI storage protocol types.
VASA Provider	Select a VASA provider name. The VASA Providers list is determined by the selected VASA protocol type.
Storage Container	Select a storage container. The Storage Containers list is determined by the selected VASA Provider. The list does not display storage containers that are already in use by other clusters.
VASA User	Select a VASA user. The VASA User list is determined by the selected VASA Provider.
Datastore Name	Enter a datastore name. The datastore name must contain 3-80 characters and cannot contain any leading or trailing whitespace.

16. Select the ESX hosts to use for creating the workload domain and click **Next**.

The ESX hosts must be commissioned with the same storage type as the primary cluster of the workload domain. For example, select hosts commissioned for vSAN ESA storage for a vSAN ESA workload domain.

17. Enter the Distributed Switch details and click **Next**.

Option	Description
Default	This profile is recommended. It provides a unified fabric for all traffic types using a single vSphere Distributed Switch.
Storage Traffic Separation	This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for storage traffic, while the other is used for all other traffic.
NSX Traffic Separation	This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for NSX (Virtual Networking) traffic, while the other is used for all other traffic.
Storage Traffic and NSX Traffic Separation	This profile creates three vSphere Distributed Switches with separate physical NICs. One switch is dedicated for NSX (Virtual Networking) traffic, the second switch is dedicated for storage traffic, while the third is used for all other traffic.
Custom Switch Configuration	Copy from a preconfigured profile or create a new distributed switch configuration. Multiple vSphere Distributed Switches can be configured. Each vSphere Distributed Switch can hold one or more network traffic configurations. Some network traffic types are mandatory. Switch configuration is incomplete until these mandatory traffic types are configured.

Option	Description
	Management, vMotion, vSAN and NSX network types can be configured only once for a cluster.
	NSX and Public network traffic types can be configured in multiple vSphere Distributed Switches.

18. Enter the vSphere Supervisor details and click **Next**.

Note: This screen is not displayed if you deactivated the **Enable vSphere Supervisor** toggle.

For more information about vSphere Supervisor, click the **Learn More** link.

Option	Description
Supervisor Name	Enter a Supervisor name.
Service CIDR	Enter an IP range for Kubernetes cluster services. The range must not overlap with any Supervisor-routable networks.
Use ESXi Management VMK settings	Select the check box to use the ESX Management VMK VLAN ID, subnet mask, gateway, and vSphere Distributed Switch for the management network.
VLAN ID	Enter a VLAN ID for the management network. This option does not appear if you selected Use ESXi Management VMK settings .
Control Plane IP Range	Enter an IP range with minimum 5 addresses for control plane VMs and management traffic.
Subnet Mask	Enter a network mask to define the IP range size. This option does not appear if you selected Use ESXi Management VMK settings . For example: 255.255.255.0
Gateway	Enter a gateway IP for external communication. This option does not appear if you selected Use ESXi Management VMK settings .
VDS	Select a vSphere Distributed Switch for the management network. This option does not appear if you selected Use ESXi Management VMK settings .
NSX Project	Select an NSX project for system and namespace VPC deployment. Note: This option is not configurable if you are deploying a new NSX Manager instance.
VPC Connectivity Profile	Select a profile for configuring VPC transit gateway connections. Note: This option is not configurable if you are deploying a new NSX Manager instance.
Private (Transit Gateway) CIDR	Enter an IP range for the private transit gateway.
Private CIDR	Enter a non-overlapping IP range for private subnet allocation.
Workload DNS	Enter DNS server addresses for all workload VMs.
Workload NTP	Enter routable NTP server addresses for workload time synchronization.

19. Review the information about the workload domain and click **Finish**.

After successful validation, workload domain creation starts.

20. Click **Fleet Management > Tasks** to monitor progress.

If you enabled vSphere Supervisor, you must deploy an NSX Edge cluster with an active-standby tier-0 gateway to complete Supervisor activation, unless you are using an existing NSX Manager instance with a compatible NSX Edge cluster. See [Configure Centralized Network Connectivity with Edge Clusters](#) for information about deploying an NSX Edge cluster.

Create a Workload Domain by Using the VCF Operations API

A workload domain adds a logical pool of compute, network, and storage infrastructure to a VCF instance. You use the VCF Operations API to customize the deployment specification by using a JSON file.

For example, the deployment wizard assumes that your ESX hosts share the same root password. If your ESX hosts have different root passwords, use the JSON specification file to deploy your environment.

In addition, if you are deploying multiple workload domains, you may find it easier to start with an existing JSON specification file and modify it for additional deployments.

The API provides additional options which are not yet available in the UI. Certain workload domain configurations, such as LACP networking or layer 3 multi-rack vSAN HCI cluster, must be deployed through the API.

Create a Workload Domain with LACP Networking

You use the VCF Operations API to create a workload domain with LACP networking

The physical network infrastructure in your environment must meet certain requirements for deployment of a workload domain, configured to use LACP on the vSphere Distributed Switch.

For each host on which you want to use LACP, you must create a separate LACP port channel on the physical switches. You must consider the following requirements when configuring LACP on the physical switches:

- The hashing algorithm of the LACP port channel on the physical switches must be the same as the hashing algorithm that is configured for the LAG on the distributed switch.
- All physical NICs that you want to connect to the LACP port channel must be configured with the same speed and duplex.
- Determine the LACP mode on the physical switch, either Active or Passive
- If the Physical switch is set to Passive mode the vDS LAG will need to be set to Active
- If the Physical switch is set Active mode the vDS LAG can be either Active or Passive

Important:

Please contact the physical switch vendor for any specific configuration required to implement LACP on the switch.

- Create a JSON specification file in a text editor.

The following example is for an environment with a single vSphere Distributed Switch. If you have multiple vSphere Distributed Switches, see the [SDDC Manager API Reference Guide](#) for details about creating a JSON specification.

```
{
  "domainName": "sfo-w01",
  "orgName": "Rainpole",
  "vcenterSpec": {
    "name": "sfo-w01-vc01",
    "networkDetailsSpec": {
      "ipAddress": "10.11.10.160",
      "dnsName": "sfo-w01-vc01.sfo.rainpole.io",
```

```

    "gateway": "10.11.10.1",
    "subnetMask": "255.255.255.0"
  },
  "rootPassword": "VMw@rel!VMw@rel!",
  "datacenterName": "sfo-w01-DC",
  "vmSize": "large"
},
"computeSpec": {
  "clusterSpecs": [
    {
      "name": "sfo-w01-cl01",
      "hostSpecs": [
        {
          "id": "<ESX host 1 ID>",
          "hostname": "sfo01-w01-r01-esx01.sfo.rainpole.io",
          "hostNetworkSpec": {
            "vmNics": [
              {
                "id": "vmnic0",
                "vdsName": "sfo-w01-cl01-vds01",
                "uplink": "uplink1"
              },
              {
                "id": "vmnic1",
                "vdsName": "sfo-w01-cl01-vds01",
                "uplink": "uplink2"
              }
            ]
          }
        },
        {
          "id": "<ESX host 2 ID>",
          "hostname": "sfo01-w01-r01-esx02.sfo.rainpole.io",
          "hostNetworkSpec": {
            "vmNics": [
              {
                "id": "vmnic0",
                "vdsName": "sfo-w01-cl01-vds01",
                "uplink": "uplink1"
              },
              {
                "id": "vmnic1",
                "vdsName": "sfo-w01-cl01-vds01",
                "uplink": "uplink2"
              }
            ]
          }
        }
      ]
    },
    {
      "id": "<ESX host 3 ID>",
      "hostname": "sfo01-w01-r01-esx03.sfo.rainpole.io",
      "hostNetworkSpec": {
        "vmNics": [

```

```

    {
      "id": "vmnic0",
      "vdsName": "sfo-w01-cl01-vds01",
      "uplink": "uplink1"
    },
    {
      "id": "vmnic1",
      "vdsName": "sfo-w01-cl01-vds01",
      "uplink": "uplink2"
    }
  ]
}
},
"datastoreSpec": {
  "vsanDatastoreSpec": {
    "datastoreName": "sfo-w01-cl01-vsan01",
    "esaConfig": {
      "enabled": "true"
    }
  }
},
"networkSpec": {
  "vdsSpecs": [
    {
      "name": "sfo-w01-cl01-vds01",
      "mtu": 8900,
      "portGroupSpecs": [
        {
          "name": "sfo-w01-cl01-vds01-pg-esxi-mgmt",
          "transportType": "MANAGEMENT",
          "standByUplinks": [],
          "teamingPolicy": "failover_explicit",
          "activeUplinks": [
            "lag"
          ]
        },
        {
          "name": "sfo-w01-cl01-vds01-pg-vmotion",
          "transportType": "VMOTION",
          "standByUplinks": [],
          "teamingPolicy": "failover_explicit",
          "activeUplinks": [
            "lag"
          ]
        }
      ]
    },
    {
      "name": "sfo-w01-cl01-vds02-pg-vsan",
      "transportType": "VSAN",
      "standByUplinks": [],
      "teamingPolicy": "failover_explicit",
      "activeUplinks": [
        "lag"
      ]
    }
  ]
}

```

```

    ]
  },
  "nsxtSwitchConfig": {
    "hostSwitchOperationalMode": "STANDARD",
    "transportZones": [
      {
        "name": "nsx-vlan-transportzone",
        "transportType": "VLAN"
      },
      {
        "name": "overlay-tz-sfo-w01-nsx01",
        "transportType": "OVERLAY"
      }
    ]
  },
  "lagSpecs": [
    {
      "name": "lag",
      "uplinksCount": 2,
      "lacpMode": "ACTIVE",
      "loadBalancingMode": "SOURCE_AND_DESTINATION_IP_AND_TCP_UDP_PORT_AND_VLAN",
      "lacpTimeoutMode": "SLOW"
    }
  ],
  "networkProfiles": [
    {
      "isDefault": true,
      "name": "sfo-w01-sfo-w01-cl01-r01-network-profile",
      "nsxtHostSwitchConfigs": [
        {
          "ipAddressPoolName": "sfo-w01-r01-ip-pool01-host",
          "uplinkProfileName": "sfo-w01-r01-uplink-profile01",
          "vdsName": "sfo-w01-cl01-vds01",
          "vdsUplinkToNsxUplink": [
            {
              "nsxUplinkName": "uplink1",
              "vdsUplinkName": "uplink1"
            },
            {
              "nsxUplinkName": "uplink2",
              "vdsUplinkName": "uplink2"
            }
          ]
        }
      ]
    }
  ],
  "nsxClusterSpec": {
    "nsxTClusterSpec": {
      "ipAddressPoolsSpec": [

```

```

    {
      "name": "sfo-w01-r01-ip-pool01-host",
      "subnets": [
        {
          "cidr": "10.13.14.0/24",
          "gateway": "10.13.14.1",
          "ipAddressPoolRanges": [
            {
              "start": "10.13.14.101",
              "end": "10.13.14.132"
            }
          ]
        }
      ]
    }
  ],
  "uplinkProfiles": [
    {
      "name": "sfo-w01-r01-uplink-profile01",
      "transportVlan": 1314,
      "teamings": [
        {
          "name": "DEFAULT",
          "policy": "FAILOVER_ORDER",
          "standByUplinks": [],
          "activeUplinks": [
            "lag"
          ]
        }
      ]
    }
  ],
  "clusterImageId": "<personality ID>"
},
"nsxTSpec": {
  "formFactor": "MEDIUM",
  "nsxManagerSpecs": [
    {
      "name": "sfo-w01-nsx01a",
      "networkDetailsSpec": {
        "ipAddress": "10.11.10.132",
        "dnsName": "sfo-w01-nsx01a.sfo.rainpole.io",
        "gateway": "10.11.10.1",
        "subnetMask": "255.255.255.0"
      }
    },
    {
      "name": "sfo-w01-nsx01b",

```

```

    "networkDetailsSpec": {
      "ipAddress": "10.11.10.133",
      "dnsName": "sfo-w01-nsx01b.sfo.rainpole.io",
      "gateway": "10.11.10.1",
      "subnetMask": "255.255.255.0"
    }
  },
  {
    "name": "sfo-w01-nsx01c",
    "networkDetailsSpec": {
      "ipAddress": "10.11.10.134",
      "dnsName": "sfo-w01-nsx01c.sfo.rainpole.io",
      "gateway": "10.11.10.1",
      "subnetMask": "255.255.255.0"
    }
  }
],
"vip": "10.11.10.131",
"vipFqdn": "sfo-w01-nsx01.sfo.rainpole.io",
"nsxManagerAdminPassword": "VMw@rel!VMw@rel!"
},
"deployWithoutLicenseKeys": "true",
"ssoDomainSpec": {
  "ssoDomainName": "sfo-w01.local",
  "ssoDomainPassword": "VMw@rel!VMw@rel!"
}
}
}

```

- In the navigation pane of the VCF Operation UI click **Developer Center > APIs & SDKs > API Explorer**
- Click the VCF Instance in which you want to create the new workload domain,
- Retrieve the unique IDs for each ESX host that are part of the stretched cluster and add the IDs to the JSON specification file.
 - Navigate to **Hosts > GET /v1/hosts**.
 - In the **Status** text box, enter `UNASSIGNED_USEABLE` and click **Execute**.
 - In the **Response** section, click **PageOfHost**, copy the `id` element of each host, and replace the respective value in the JSON specification file.

ESX Host	Value
ESX Host 1	<i>ESX host 1 ID</i>
ESX Host 2	<i>ESX host 2 ID</i>
ESX Host 3	<i>ESX host 3 ID</i>

- Retrieve the vSphere Lifecycle Manager image ID to use for the cluster and add the ID to the JSON specification file.
 - Navigate to **Personalities > GET /v1/personalities** and click **Execute**.
 - In the **Response** section, click the image.
 - Copy the `personalityId` element and replace the `<personality ID>` value in the JSON specification file.
- Expand the **Domains** API category.
- To validate the JSON specification, paste the JSON specification in the body of the `POST /v1/domains/validations` API and click **Execute**.
- To create the workload domain, paste the JSON specification in the body of the `POST /v1/domains/` API and click **Execute**.
- Click **View Task Status** to view the domain creation tasks and sub tasks.

Create and Configure a Workload Domain with a Layer 3 Multi-Rack vSAN HCI Cluster

A workload domain with a layer 3 multi-rack vSAN HCI cluster provides scalability, resilience, and optimal resource usage for workloads.

The clusters in such workload domains have the following configuration:

- Layer 3 network fabric between the racks.
- A dedicated NSX host sub-transport node profile for each rack.
- Each rack can be configured as a vSAN fault domain

Deploy the L3 multi-rack vSAN workload domain so ESX hosts in different racks communicate over the L3 network and vSAN spans those racks. You then disable the vSAN automatic policy management, set and assign a default vSAN storage policy to match availability targets, and create vSAN fault domains to map racks and protect against rack-level failure.

For more information on the layer 3 multi-rack cluster design, see the *VMware Cloud Foundation Design Guide*.

Create a Workload Domain with a Layer 3 Multi-Rack vSAN HCI Cluster

Deploying a workload domain with a layer 3 vSAN HCI cluster that spans multiple racks requires routed networks between racks. To deploy such a workload domain, you use the VCF API and continue the setup after the deployment is complete.

- Create a JSON specification file in a text editor.

The following example is for an environment with a single vSphere Distributed Switch. If you have multiple vSphere Distributed Switches, see the [SDDC Manager API Reference Guide](#) for details about creating a JSON specification.

```

{
  "domainName": "sfo-w01",
  "vcenterSpec": {
    "name": "sfo-w01-vc01",
    "networkDetailsSpec": {
      "ipAddress": "10.11.10.160",
      "dnsName": "sfo-w01-vc01.sfo.rainpole.io"
    }
  },
  "rootPassword": "VMw@rel!VMw@rel!",
  "datacenterName": "sfo-w01-dc01",
  "vmSize": "Medium"
},
"computeSpec": {
  "clusterSpecs": [
    {

```

```

"name": "sfo-w01-cl01",
"hostSpecs": [
{
  "id": "<ESX host 1 ID>",
  "hostname": "sfo01-w01-r01-esx01.sfo.rainpole.io",
  "hostNetworkSpec": {
    "vmNics": [
      {
        "id": "vmnic0",
        "vdsName": "sfo-w01-cl01-vds01",
        "uplink": "uplink1"
      },
      {
        "id": "vmnic1",
        "vdsName": "sfo-w01-cl01-vds01",
        "uplink": "uplink2"
      }
    ],
    "networkProfileName": "sfo01-w01-r01-network-profile01"
  }
},
{
  "id": "<ESX host 2 ID>",
  "hostname": "sfo01-w01-r02-esx01.sfo.rainpole.io",
  "hostNetworkSpec": {
    "vmNics": [
      {
        "id": "vmnic0",
        "vdsName": "sfo-w01-cl01-vds01",
        "uplink": "uplink1"
      },
      {
        "id": "vmnic1",
        "vdsName": "sfo-w01-cl01-vds01",
        "uplink": "uplink2"
      }
    ],
    "networkProfileName": "sfo01-w01-r02-network-profile01"
  }
},
{
  "id": "<ESX host 3 ID>",
  "hostname": "sfo01-w01-r03-esx01.sfo.rainpole.io",
  "hostNetworkSpec": {
    "vmNics": [
      {
        "id": "vmnic0",
        "vdsName": "sfo-w01-cl01-vds01",
        "uplink": "uplink1"
      },
      {
        "id": "vmnic1",
        "vdsName": "sfo-w01-cl01-vds01",
        "uplink": "uplink2"
      }
    ],
    "networkProfileName": "sfo01-w01-r03-network-profile01"
  }
},
{
  "id": "<ESX host 4 ID>",
  "hostname": "sfo01-w01-r04-esx01.sfo.rainpole.io",
  "hostNetworkSpec": {
    "vmNics": [
      {
        "id": "vmnic0",
        "vdsName": "sfo-w01-cl01-vds01",
        "uplink": "uplink1"
      },
      {
        "id": "vmnic1",
        "vdsName": "sfo-w01-cl01-vds01",
        "uplink": "uplink2"
      }
    ],
    "networkProfileName": "sfo01-w01-r04-network-profile01"
  }
},
{
  "datastoreSpec": {
    "vsanDatastoreSpec": {
      "datastoreName": "sfo-w01-cl01-vsan01",
      "esaConfig": {
        "enabled": "true"
      }
    }
  },
  "networkSpec": {
    "vdsSpecs": [
      {
        "name": "sfo-w01-cl01-vds01",
        "mtu": 9000,
        "portGroupSpecs": [
          {
            "name": "sfo-w01-cl01-vds01-pg-esxi-mgmt",
            "transportType": "MANAGEMENT",
            "standByUplinks": [],
            "teamingPolicy": "loadbalance_loadbased",
            "activeUplinks": [
              "uplink1",
              "uplink2"
            ]
          },
          {
            "name": "sfo-w01-cl01-vds01-pg-vmotion",
            "transportType": "VMOTION",

```

```

        "uplink": "uplink2"
      }
    ],
    "networkProfileName": "sfo01-w01-r03-network-profile01"
  }
},
{
  "id": "<ESX host 4 ID>",
  "hostname": "sfo01-w01-r04-esx01.sfo.rainpole.io",
  "hostNetworkSpec": {
    "vmNics": [
      {
        "id": "vmnic0",
        "vdsName": "sfo-w01-cl01-vds01",
        "uplink": "uplink1"
      },
      {
        "id": "vmnic1",
        "vdsName": "sfo-w01-cl01-vds01",
        "uplink": "uplink2"
      }
    ],
    "networkProfileName": "sfo01-w01-r04-network-profile01"
  }
},
{
  "datastoreSpec": {
    "vsanDatastoreSpec": {
      "datastoreName": "sfo-w01-cl01-vsan01",
      "esaConfig": {
        "enabled": "true"
      }
    }
  },
  "networkSpec": {
    "vdsSpecs": [
      {
        "name": "sfo-w01-cl01-vds01",
        "mtu": 9000,
        "portGroupSpecs": [
          {
            "name": "sfo-w01-cl01-vds01-pg-esxi-mgmt",
            "transportType": "MANAGEMENT",
            "standByUplinks": [],
            "teamingPolicy": "loadbalance_loadbased",
            "activeUplinks": [
              "uplink1",
              "uplink2"
            ]
          },
          {
            "name": "sfo-w01-cl01-vds01-pg-vmotion",
            "transportType": "VMOTION",

```

```

        "standByUplinks": [],
        "teamingPolicy": "loadbalance_loadbased",
        "activeUplinks": [
            "uplink1",
            "uplink2"
        ]
    },
    {
        "name": "sfo-w01-cl01-vds01-pg-vsan",
        "transportType": "VSAN",
        "standByUplinks": [],
        "teamingPolicy": "loadbalance_loadbased",
        "activeUplinks": [
            "uplink1",
            "uplink2"
        ]
    }
],
"nsxtSwitchConfig": {
    "hostSwitchOperationalMode": "STANDARD",
    "transportZones": [
        {
            "name": "vlan-tz-sfo-w01-nsx01",
            "transportType": "VLAN"
        },
        {
            "name": "overlay-tz-sfo-w01-nsx01",
            "transportType": "OVERLAY"
        }
    ]
}
],
"networkProfiles": [
    {
        "isDefault": true,
        "name": "sfo01-w01-r01-network-profile01",
        "nsxtHostSwitchConfigs": [
            {
                "ipAddressPoolName": "sfo01-w01-r01-ip-pool01-host",
                "uplinkProfileName": "sfo01-w01-r01-uplink-profile01",
                "vdsName": "sfo-w01-cl01-vds01",
                "vdsUplinkToNsxUplink": [
                    {
                        "nsxUplinkName": "uplink1",
                        "vdsUplinkName": "uplink1"
                    },
                    {
                        "nsxUplinkName": "uplink2",
                        "vdsUplinkName": "uplink2"
                    }
                ]
            }
        ]
    }
]
}

```

```

    ]
},
{
    "isDefault": false,
    "name": "sfo01-w01-r02-network-profile01",
    "nsxtHostSwitchConfigs": [
        {
            "ipAddressPoolName": "sfo01-w01-r02-ip-pool01-host",
            "uplinkProfileName": "sfo01-w01-r02-uplink-profile01",
            "vdsName": "sfo-w01-cl01-vds01",
            "vdsUplinkToNsxUplink": [
                {
                    "nsxUplinkName": "uplink1",
                    "vdsUplinkName": "uplink1"
                },
                {
                    "nsxUplinkName": "uplink2",
                    "vdsUplinkName": "uplink2"
                }
            ]
        }
    ]
},
{
    "isDefault": false,
    "name": "sfo01-w01-r03-network-profile01",
    "nsxtHostSwitchConfigs": [
        {
            "ipAddressPoolName": "sfo01-w01-r03-ip-pool01-host",
            "uplinkProfileName": "sfo01-w01-r03-uplink-profile01",
            "vdsName": "sfo-w01-cl01-vds01",
            "vdsUplinkToNsxUplink": [
                {
                    "nsxUplinkName": "uplink1",
                    "vdsUplinkName": "uplink1"
                },
                {
                    "nsxUplinkName": "uplink2",
                    "vdsUplinkName": "uplink2"
                }
            ]
        }
    ]
},
{
    "isDefault": false,
    "name": "sfo01-w01-r04-network-profile01",
    "nsxtHostSwitchConfigs": [
        {
            "ipAddressPoolName": "sfo01-w01-r04-ip-pool01-host",
            "uplinkProfileName": "sfo01-w01-r04-uplink-profile01",
            "vdsName": "sfo-w01-cl01-vds01",
            "vdsUplinkToNsxUplink": [

```



```

        {
            "nsxUplinkName": "uplink1",
            "vdsUplinkName": "uplink1"
        },
        {
            "nsxUplinkName": "uplink2",
            "vdsUplinkName": "uplink2"
        }
    ]
}
]
},
"nsxClusterSpec": {
    "nsxTClusterSpec": {
        "ipAddressPoolsSpec": [
            {
                "name": "sfo01-w01-r01-ip-pool01-host",
                "subnets": [
                    {
                        "cidr": "10.15.14.0/24",
                        "gateway": "10.15.14.1",
                        "ipAddressPoolRanges": [
                            {
                                "start": "10.15.14.101",
                                "end": "10.15.14.132"
                            }
                        ]
                    }
                ]
            }
        ]
    },
    {
        "name": "sfo01-w01-r02-ip-pool01-host",
        "subnets": [
            {
                "cidr": "10.15.25.0/24",
                "gateway": "10.15.25.1",
                "ipAddressPoolRanges": [
                    {
                        "start": "10.15.25.101",
                        "end": "10.15.25.116"
                    }
                ]
            }
        ]
    }
],
},
{
    "name": "sfo01-w01-r03-ip-pool01-host",
    "subnets": [
        {
            "cidr": "10.15.36.0/24",
            "gateway": "10.15.36.1",
            "ipAddressPoolRanges": [

```

```

        {
            "start": "10.15.36.101",
            "end": "10.15.36.116"
        }
    ]
}
]
},
{
    "name": "sfo01-w01-r04-ip-pool01-host",
    "subnets": [
        {
            "cidr": "10.15.47.0/24",
            "gateway": "10.15.47.1",
            "ipAddressPoolRanges": [
                {
                    "start": "10.15.47.101",
                    "end": "10.15.47.132"
                }
            ]
        }
    ]
}
],
"uplinkProfiles": [
    {
        "name": "sfo01-w01-r01-uplink-profile01",
        "transportVlan": 1514,
        "teamings": [
            {
                "name": "DEFAULT",
                "policy": "LOADBALANCE_SRCID",
                "standByUplinks": [],
                "activeUplinks": [
                    "uplink1",
                    "uplink2"
                ]
            }
        ]
    }
],
{
    "name": "sfo01-w01-r02-uplink-profile01",
    "transportVlan": 1525,
    "teamings": [
        {
            "name": "DEFAULT",
            "policy": "LOADBALANCE_SRCID",
            "standByUplinks": [],
            "activeUplinks": [
                "uplink1",
                "uplink2"
            ]
        }
    ]
}
]
},
{
    "name": "sfo01-w01-r03-uplink-profile01",
    "transportVlan": 1525,
    "teamings": [
        {
            "name": "DEFAULT",
            "policy": "LOADBALANCE_SRCID",
            "standByUplinks": [],
            "activeUplinks": [
                "uplink1",
                "uplink2"
            ]
        }
    ]
}
]
},
{
    "name": "sfo01-w01-r04-uplink-profile01",
    "transportVlan": 1525,
    "teamings": [
        {
            "name": "DEFAULT",
            "policy": "LOADBALANCE_SRCID",
            "standByUplinks": [],
            "activeUplinks": [
                "uplink1",
                "uplink2"
            ]
        }
    ]
}
]
},
{
    "name": "sfo01-w01-r05-uplink-profile01",
    "transportVlan": 1525,
    "teamings": [
        {
            "name": "DEFAULT",
            "policy": "LOADBALANCE_SRCID",
            "standByUplinks": [],
            "activeUplinks": [
                "uplink1",
                "uplink2"
            ]
        }
    ]
}
]
}
]
}

```

```
    ]
  },
  {
    "name": "sfo01-w01-r03-uplink-profile01",
    "transportVlan": 1536,
    "teamings": [
      {
        "name": "DEFAULT",
        "policy": "LOADBALANCE_SRCID",
        "standByUplinks": [],
        "activeUplinks": [
          "uplink1",
          "uplink2"
        ]
      }
    ]
  }
],
{
  "name": "sfo01-w01-r04-uplink-profile01",
  "transportVlan": 1547,
  "teamings": [
    {
      "name": "DEFAULT",
      "policy": "LOADBALANCE_SRCID",
      "standByUplinks": [],
      "activeUplinks": [
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        "uplink2"
      ]
    }
  ]
}
]
}
}
},
"clusterImageId": "<personality ID>"
}
]
},
"nsxTSpec": {
  "formFactor": "Large",
  "nsxManagerSpecs": [
    {
      "name": "sfo-w01-nsx01a",
      "networkDetailsSpec": {
        "ipAddress": "10.11.10.162",
        "dnsName": "sfo-w01-nsx01a.sfo.rainpole.io"
      }
    }
  ],
  "vip": "10.11.10.161",
  "vipFqdn": "sfo-w01-nsx01.sfo.rainpole.io",
```

```
    "nsxManagerAdminPassword": "VMw@rel!VMw@rel!"
  },
  "ssoDomainSpec": {
    "ssoDomainName": "sfo-w01.local",
    "ssoDomainPassword": "VMw@rel!VMw@rel!"
  },
  "deployWithoutLicenseKeys": "true"
}
```

2. In the navigation pane of the VCF Operation UI click **Developer Center > APIs & SDKs > API Explorer**
3. Click the VCF Instance in which you want to create the layer 3 multi-rack vSAN HCI cluster.
4. Retrieve the unique IDs for each ESX host that will be part of the stretched cluster and add the IDs to the JSON specification file.
 - a) Navigate to **Hosts > GET /v1/hosts**.
 - b) In the **Status** text box, enter UNASSIGNED_USEABLE and click **Execute**.
 - c) In the **Response** section, click **PageOfHost**, copy the `id` element of each host, and replace the respective value in the JSON specification file.

ESX Host	Value
ESX Host 1	<i>ESX host 1 ID</i>
ESX Host 2	<i>ESX host 2 ID</i>
ESX Host 3	<i>ESX host 3 ID</i>
ESX Host 4	<i>ESX host 4 ID</i>

5. Retrieve the vSphere Lifecycle Manager image ID to use for the cluster and add the ID to the JSON specification file. Navigate to **Personalities > GET /v1/personalities**. Click **Execute**. In the **Response** section, click the personality representing the image you want to use, copy the `personalityId` element, and replace the *<personality ID>* value in the JSON specification file.
6. Expand the **Domains** API category.
7. To validate the JSON specification, paste the JSON specification in the body of the POST `/v1/domains/validations` API and click **Execute**.
8. To create the workload domain, paste the JSON specification in the body of the POST `/v1/domains/` API and click **Execute**.
9. Click **View Task Status** to view the domain creation tasks and sub tasks.

Deactivate Auto Policy Management for vSAN ESA Cluster

You cannot use auto policy management in a multi-rack cluster when using vSAN fault domains. This must be disabled if vSAN fault domains are configured for each rack.

1. In a Web browser, log in to the workload domain vCenter at `https://<vcenter_server_fqdn>/ui` as **administrator@vsphere.local**.
2. In the **Hosts and clusters** inventory, navigate to the multi-rack vSAN cluster.
3. On the **Configure** tab for the cluster, select **vSAN > Services**.
4. In the **vSAN ESA** pane, click **Edit**.
5. Toggle **Auto-policy Management** off and click **Apply**.

Create and Assign a New Default Storage Policy for vSAN ESA Cluster

By default, the vSAN ESA default storage policy is based on the number of hosts in the cluster during creation. To provide for fault domains that allow rack resiliency in a multi-rack cluster, you must create a default storage policy that is compatible with the number of racks.

1. In the vSphere Client for the workload domain vCenter, from the vSphere Client menu, select **Policies and Profiles**.
2. On the **VM Storage Policies** page, click **Create**.
3. On the **Name and description** page, select the vCenter instance that manages the multi-rack cluster.
4. Type a name and a description for the storage policy and click **Next**.
5. On the **Policy structure** page, select **Enable rules for "vSAN" storage**, and click **Next**.
6. On the **Availability** tab of the **vSAN** page, set **Failures to tolerate** according to the configuration of your environment. You use RAID 5 or RAID 6 (erasure coding) for vSAN ESA.
7. On the **Storage Rules** tab, select **All flash** and click **Next**.
8. On the **Storage compatibility** page, verify that the datastore is compatible and click **Next**.
9. On the **Review and finish** page, review the policy settings and click **Finish**.
10. In the **Storage** inventory of the vSphere Client, navigate to the vSAN datastore.
11. On the **Configure** tab, click **General** and next to the Default Storage Policy pane, click **Edit**.
12. Select the storage policy that you want to assign as default to the vSAN datastore and click **OK**.

Create vSAN Fault Domains

For vSAN multi-rack clusters with more than one ESX node per rack create a vSAN fault domain for each rack for rack resilience.

1. In the vSphere Client for the workload domain vCenter, in the **Hosts and clusters** inventory, navigate to the multi-rack cluster.
2. On the **Configure** tab, select **vSAN > Fault Domains**.
3. Click the plus icon.
4. In the **New Fault Domain** dialog box, enter the fault domain name, and select the hosts for the fault domain.
5. Click **Create**.
6. Repeat these steps for the other racks.

During a future cluster expansion, add hosts manually to their respective fault domains.

Import an Existing vCenter to Create a Workload Domain

If you have an existing vCenter instance, with or without NSX Manager, you can use VCF Operations to import it to a VCF instance as a workload domain.

- Verify that your existing vSphere infrastructure satisfies the requirements and supported configurations that are listed below.
- Verify that SSH is enabled on the existing vCenter instance.

Note:

All vSphere clusters in the existing vCenter are imported as part of this process. You cannot select a subset of clusters.

Minimum Component Versions

- VCF 9.0.x instance from which to perform the import
- vCenter 8.0 Update 1 or later
- ESX 8.0 Update 1 or later
- NSX Manager 4.1.0.2 or later

Note:

If NSX Manager is not present, you deploy a new instance during the import procedure. The version of the deployed NSX is the latest compatible NSX version for the related vCenter instance. To change the NSX version, see [KB 429205](#).

After the import, you can only upgrade your VCF environment to a version that is released after the release date of the earliest version of the imported VCF components. If the imported NSX version is released after a specific VCF minor or maintenance release, this workload domain cannot be upgraded to that specific VCF minor or maintenance version. See [VMware Interoperability Matrix](#).

The import workflow automatically activates NSX on Distributed Virtual Port Groups (DVPGs). This action automatically activates the Distributed Firewall (DFW) on all DVPGs in the imported cluster, applying firewall rules to existing workloads. The default DFW rules allow all Layer 2 and Layer 3 traffic. After the import completes, you can deactivate NSX on DVPGs by using the NSX UI or the Transport Node Collection API.

The following table lists supported and not supported configurations for importing your existing infrastructure to a workload domain.

Table 822: Supported Configurations

Category	Supported	Not Supported
Storage	• Enough free space for the full deployment of VMware Cloud Foundation, according to the <i>Planning and Preparation Workbook</i> • Datastore shared across all hosts in the cluster, accessible from all hosts in the cluster, and writable by all hosts in the cluster	

Category	Supported	Not Supported
	- Any supported vSphere storage type. If the default cluster includes multiple datastore types, VCF determines the primary datastore using the following priority: - vSAN - NFS v3 - VMFS - NFS 4.1 - iSCSI - Vols* Note: If you are using a <code>JSON</code> specification file to converge, you can use the <code>existingDatastoreName</code> property to override the priority order and use a specific datastore. - vSAN Stretched Clusters (ESA, OSA, or ESA storage clusters) with at least 3 ESX hosts in each availability zone and a witness host deployed in a vSAN witness zone that is different from the availability zones - Two-node vSAN clusters for remote offices/branch offices (ROBO) with two ESX hosts and one witness host for VCF simple deployments Note: For two-node vSAN clusters, the admission control policy should be configured to $\geq 50\%$ for both memory and CPU. - vSAN OSA clusters, where deduplication and compression are either both activated or both deactivated - vSAN OSA clusters with activated compression only	
Network	- Virtual Distributed Switch (VDS) 8.0 or later Note: You can move to a vSphere Distributed Switch. See Migrating from Standard to Distributed vSwitch. - Ports aligned with VMware Ports and Protocols - vCenter instance with temporary IP address - Statically assigned VMkernel IP addresses, except NSX Host TEPs which can include DHCP assigned addresses - Dedicated network for VMware vMotion - vSphere Distributed Switches (VDS) with enabled link aggregation control protocol (LACP) - ESX hosts with a combination of standard switches and VDS - ESX hosts with multiple physical uplinks of minimum of 10Gbps, that are assigned to a vSphere Distributed Switch	- Cisco virtual switches - vCenter instances without a vSphere Distributed Switch - Custom vCenter ports for client connections - Environments using ports not aligned with VMware Ports and Protocols - Dynamically allocated VMkernel IP addresses Note: You can move to statically assigned IP addresses. See Edit a VMkernel Adapter Configuration. - Shared networks for VMware vMotion - Multiple NSX Manager instances managing a single vCenter instance - NSX Bare Metal Edge nodes

Category	Supported	Not Supported
	- ESX hosts with a single pNIC with a minimum of 10 Gbps speed, that is connected to a standard switch - Clusters using shared vSphere Distributed Switches - NSX Manager instance that is shared across multiple vCenter instances without ELM Note: If no NSX Manager instance is connected to the vCenter environment, during the import operation, a cluster of 3x NSX Managers is deployed. - NSX Manager instance that is on the same network segment as the imported vCenter appliance without ELM - Starting with VCF 9.0.1, vCenter instances without ELM and with existing NSX Local Managers that are part of an NSX Federation configuration - ESX hosts with a single Physical Network Adapter (pNIC) - DNS server with forward and reverse DNS records - DNS server must be configured for all management domain components - All FQDNs used for the different components must be resolvable to a unique IP - vCenter instances and NSX Managers that are located on different network segments	
Compute	- Virtual machine of vCenter instance hosted on one of the clusters that is managed by the same instance Note: You can move the virtual machine to a supported location by using VMware vMotion. See Migrating Virtual Machines Between vCenter Systems. - Virtual machine of vCenter instance hosted on the default cluster in the management domain - Clusters using vSphere Lifecycle Manager images Note: You can convert clusters that use baselines to use images before upgrading to ESX 9.0. See Convert a Cluster or a Host That Uses Baselines Into a Cluster or a Host That Uses vSphere Lifecycle Manager Images. - Clusters using fully automated Distributed Resource Scheduler (DRS) - Standalone ESX hosts and single ESX host clusters, if your vCenter instance has an additional cluster that meets the cluster requirements Note: You can manually perform the lifecycle management of the standalone ESX hosts and single ESX host clusters by using the vSphere Client.	- Clusters managed by Dell VxRail - Clusters using vSphere Configuration Profiles - vCenter instances with ELM Note: If your vCenter instance does not have existing NSX registrations, you can manually upgrade your vCenter instances to version 9.0.x and then break ELM for all vCenter instances. See Deactivate Enhanced Link Mode from vCenter Using the cmsso-util break-elm Utility. - vCenter instances with a shared NSX Manager in different VCF instances. - Clusters using baselines for lifecycle management - Clusters using manual or partially automated Distributed Resource Scheduler (DRS) Note: You can set the automation level to fully automated. See Edit Cluster Settings. - Partial import of clusters - All clusters managed by a vCenter instance are imported together

Category	Supported	Not Supported
	Clusters using fully automated Distributed Resource Scheduler (DRS)	

* vSphere Virtual Volumes (vVols) framework is deprecated in version 9 and using it is not recommended.

Note:

If you are importing an NSX Federation configuration, only the Local Manager is imported. To upgrade such environment, see [Upgrading NSX in a Federated Environment](#).

Some functionalities are not available for imported clusters:

- Adding hosts to imported clusters without the use of the vSphere Client
- Password Management of imported ESX hosts via the Password Management Console of VCF Operations.

Note:

You can use the workaround in the [Broadcom Knowledge Base Article 388859](#) to create the missing ESXi root credential entries in SDDC Manager to enable password management for these hosts.

- In VCF Operations, select **Inventory > Detailed View**.
- Expand **VCF Instances** and browse to the VCF Instance into which you want to import a workload domain.
- Click **Add Workload Domain > Import a vCenter**.
- Enter a workload domain name and click **Next**.
- Select a vCenter and optionally an NSX Manager instance to import as a workload domain and click **Next**.

Option	Description
Select an existing standalone vCenter from the inventory	Select an existing vCenter from the table. If the vCenter is connected to an NSX Manager instance, activate the toggle and enter the following details: - NSX Manager VIP FQDN - NSX Administrator Password - NSX Root Password - NSX Audit Password If your existing NSX Manager instance is version 9.0 or higher and has one or more Edge clusters, the import process adds the discovered Edge node VMs to the VCF inventory, including their user credentials. During the import process, the Edge node credential passwords will be reset. You can retrieve the updated passwords from the VCF credential store. Select the Enable Edge cluster sync and import NSX Edge node VMs check box to perform the password reset operation for all the nodes in the existing Edge cluster, as well as for any nodes added later.
Specify an external vCenter	Enter the following details for the external vCenter: - vCenter FQDN - vCenter Root Password - SSO user name - SSO user password If the vCenter is connected to an NSX Manager instance, activate the toggle and enter the following details: - NSX Manager VIP FQDN

Option	Description
	- NSX Administrator Password - NSX Root Password - NSX Audit Password If your existing NSX Manager instance is version 9.0 or higher and has one or more Edge clusters, the import process adds the discovered Edge node VMs to the VCF inventory, including their user credentials. During the import process, the Edge node credential passwords will be reset. You can retrieve the updated passwords from the VCF credential store. Select the Enable Edge cluster sync and import NSX Edge node VMs check box to perform the password reset operation for all the nodes in the existing Edge cluster, as well as for any nodes added later.

- Confirm the certificate thumbprints and click **Next**.
This includes vCenter, NSX Manager (if you are using an existing instance), and additional thumbprints.
- Run the prechecks and resolve any issues.
- If you are not importing an NSX Manager instance that is connected to your vCenter, enter the NSX Manager instance details and click **Next**.
Choose whether to create a new NSX Manager instance or join an existing NSX Manager instance. If you already have an NSX Manager instance for a different workload domain, you can reuse that NSX Manager instance or create a new one.
Note: You cannot join the existing NSX Manager instance for an imported or newly deployed management domain.
If you are using an existing NSX Manager instance, select it. If you are creating a new NSX Manager instance, enter the following information.

Option	Description
Deployment Size	Standard: Deploys single-node NSX Manager requiring fewer resources. You can choose a standard deployment size only when importing a vCenter 9.0 and later instance. High-Availability (recommended): Deploys a three-node NSX Manager cluster to enhance the availability of management and control plane functions, as well as minimize downtime during upgrades.
Appliance Size	- Medium - Large - Extra Large
Appliance 1 FQDN	Enter the FQDN for the first NSX Manager node.
Appliance 2 FQDN	Enter the FQDN for the second NSX Manager node.
Appliance 3 FQDN	Enter the FQDN for the third NSX Manager node. This is not applicable if you chose Standard deployment size.
Appliance Cluster FQDN	Enter the NSX Manager cluster FQDN.
Administrator Password	Enter and confirm an NSX administrator password and enable auto-generated passwords. Requirements: - At least five different characters. - At least one lowercase character.

Option	Description
	- At least one digit. - No more than four monotonic or sequential characters. - At least 12 characters. - Cannot be palindrome. - At least one special character. - At least one uppercase character. - No dictionary words.
Auditor Password	Enter and confirm an NSX administrator password or enable auto-generated passwords. Requirements: - At least five different characters. - At least one lowercase character. - At least one digit. - No more than four monotonic or sequential characters. - At least 12 characters. - Cannot be palindrome. - At least one special character. - At least one uppercase character. - No dictionary words.
NSX overlay networking	Choose whether or not to use the ESX management VMkernel for NSX overlay networking.

9. Run the validations and prechecks and click **Next**.
Resolve any issues before proceeding.

10. Review your settings and click **Finish**.

Delete a Workload Domain

You can delete a workload domain in VCF Operations.

- If remote vSAN datastores are mounted on a cluster in the workload domain, then the workload domain cannot be deleted. To delete such workload domains, you must first migrate any virtual machines from the remote datastore to the local datastore and then unmount the remote vSAN datastores from vCenter.
- If you require access after deleting a workload domain, back up the data. The datastores on the workload domain are destroyed when it is deleted.
- Migrate the virtual machines that you want to keep to another workload domain using cross vCenter vMotion.
- Delete any NSX Edge clusters hosted on the workload domain.

When you delete a workload domain, the clusters within that workload domain are deleted and the hosts are returned to the free pool with a `need_cleanup` host state.

Deleting a workload domain also removes the components associated with the workload domain from the management domain. This includes the vCenter instance and the NSX Manager cluster instances.

Note:

If the NSX Manager cluster is shared with any other workload domains, it will not be deleted.

Note:

If the workload domain shares an NSX Manager cluster and VCF Operations for networks latency collection is enabled for the NSX Manager cluster, NSX objects from the domain may be referenced and stop the workload domain deletion. Disable the latency collection and delete the workload domain.

The network pools used by the workload domain are not deleted as part of the workload domain deletion process and must be deleted separately.

CAUTION: Deleting a workload domain is an irreversible operation. All clusters and virtual machines within the workload domain are deleted and the underlying datastores are destroyed.

It can take up to 20 minutes for a workload domain to be deleted. During this process, you cannot perform any operations on workload domains.

- In VCF Operations, select **Inventory > Detailed View**.
- Expand **VCF Instances** and browse to the VCF instance that includes the workload domain you want to delete.
- Click the workload domain name and select **Delete Domain** from the **Actions** menu.
- Click **Delete Workload Domain**.
A message indicating that the workload domain is being deleted appears.

Decommission the hosts that were part of the workload domain, then re-image with ESX and commission them again. See [Managing ESX Hosts in VMware Cloud Foundation](#).

Expand a VCF Domain

You can expand the management domain or a workload domain to add resources to support additional workloads or availability.

To expand a VCF domain, you can:

- Add an ESX host from the global inventory to an SDDC cluster.
By adding an individual host to an existing VCF domain, you can expand the amount of resources contained within an existing SDDC cluster up to the supported vSphere maximums.
- Add a new SDDC cluster to a VCF domain.
VCF domains support multiple SDDC clusters. You can add an additional SDDC cluster to an existing VCF domain to provide for increased capacity.

Add an ESX Host to an SDDC Cluster in a VCF Domain

You can add an ESX host to an SDDC cluster using the vSphere Client. Adding an ESX host to an SDDC cluster in a VCF domain increases the available resources. You can add multiple hosts at the same time.

- Verify that one or more unassigned hosts are available in the Hosts inventory. See [View ESX Host Inventory](#).
- For information on commissioning hosts, see [Commission ESX Hosts](#).
- Verify that the ESX hosts you want to add is in an active state.
- Verify that the ESX host to be added matches the configuration of the ESX hosts already in the SDDC cluster. This allows the SDDC cluster configuration to remain balanced. If the host to be added does not match the pre-existing hosts in the SDDC cluster, the cluster will be unbalanced and a warning is displayed. The warning does not prevent the expansion and can be dismissed if needed.
- Verify that the ESX host you are adding has the same type of principal storage as the existing ESX hosts in the SDDC cluster. A host using NFS for principal storage will automatically use the same NFS configuration as the other hosts in the SDDC cluster. For a host using VMFS on FC, you must configure zoning, mount the associated volumes, and create the datastore on the host before adding the host to a SDDC cluster. A host using vVols for principal storage will automatically use the same vVols configuration as the other hosts in the SDDC cluster.
- If the SDDC cluster to which you are adding hosts uses a static IP pool for the NSX Host Overlay Network TEPs, that pool must include enough IP addresses for the hosts you are adding.
- Different NIC enumeration is allowed.
- To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See [Configuring VCF Single Sign-On](#) and [Linking](#)

[vCenter instances in VCF Operations](#). As an alternative, you can perform this task using the SDDC Manager UI (<https://<sddc-manager-fqdn>>).

When you use the vSphere Client to add an ESX host to an SDDC cluster, it uses the cluster's existing vDS for management, vMotion, vSAN, and host overlay traffic.

Note:

You can add hosts to or remove hosts from multiple different SDDC clusters in parallel. For example, you add three hosts to Cluster A, and while that task is running, you can start a separate task to add (or remove) four hosts to Cluster B. See [VMware Configuration Maximums](#) for information about the maximum number of add/remove hosts tasks that you can run in parallel.

1. In the vSphere Client, browse to the vSphere cluster in the inventory.
2. Select the vSphere cluster and click **Actions > Add Hosts > Add Unassigned Hosts**.
3. Select the host to add to the vSphere cluster and click **Next**.

You can view both vDS and NSX networking configuration on the cluster. Each vDS has two uplinks in it, and you can assign any free VMNICs on the host to the existing uplinks on the vDS.

4. Click **Next**.
5. Review the host, switch configuration, and license details and click **Finish**.

Add an SDDC Cluster to a VCF Domain

If you want to add capacity to a VCF domain, you can use the vSphere Client to add an SDDC cluster to a VCF domain.

- Verify that there are unassigned hosts are available in the Hosts inventory. See [View ESX Host Inventory](#).
- For information on commissioning hosts, see [Commission ESX Hosts](#).
- Ensure that the hosts you want to add to the SDDC cluster are in an active state.
- If you are using DHCP for the NSX Host Overlay Network, a DHCP server must be configured and advertising IP addresses on the NSX Host Overlay VLAN of the VCF domain. When NSX creates TEPs for the VCF domain, they are assigned IP addresses from the DHCP server.
- To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See [Configuring VCF Single Sign-On](#) and [Linking vCenter instances in VCF Operations](#). As an alternative, you can perform this task using the SDDC Manager UI (<https://<sddc-manager-fqdn>>).

If a workload domain has multiple clusters, each cluster can use a different type of principal storage, as long as all hosts within an SDDC cluster use the same type.

You can run multiple add cluster tasks at the same time.

Important:

Starting with VMware Cloud Foundation 9.0 and vSphere Foundation 9.0, the vSphere Virtual Volumes capability, also known as vVols, is deprecated and will be removed in a future release of VMware Cloud Foundation and vSphere Foundation. Support for vSphere Virtual Volumes will continue for critical bug fixes only for versions of vSphere 8.x, VMware Cloud Foundation and vSphere Foundation 5.x, and other supported versions until end-of-support for the respective release.

1. In the vSphere Client for the VCF domain vCenter, browse to the VCF instance datacenter in the inventory.
2. Select the datacenter and click **Actions > New Cluster > Create SDDC Cluster**.
3. Select the storage type for the SDDC cluster and click **Begin**.

Option	Description
vSAN	Select to configure a vSAN-based workload domain.

Option	Description
	Select Enable vSAN ESA to configure a vSAN ESA workload domain. If you do not select this option, the workload domain is configured with vSAN OSA. Note: vSAN storage clusters (formerly known as vSAN Max) require vSAN ESA
NFS	Select to configure an NFS-based workload domain.
VMFS on FC	Select to configure a Fibre Channel-based workload domain
vVol	Select to configure a vVol-based workload domain.

4. Enter a name for the SDDC cluster.
5. Select a cluster image from the drop-down menu.

Note: If the cluster image contains a different version of a vendor add-on or component than what is installed on the ESXi hosts you add to the cluster, the hosts will be remediated to use the cluster image during cluster creation.

6. Click **Next**.
7. Enter the vSAN Storage details and click **Next**.

This screen only displays if you selected vSAN storage. The vSAN storage options are different for vSAN OSA and vSAN ESA.

vSAN OSA:

Option	Description
vSAN Cluster Type	• vSAN HCI: Provides storage and compute resources. • vSAN Compute Cluster: Provides compute resources only.
Failures to Tolerate (vSAN HCI only)	Specify the level of availability you want configured for the cluster. Your selection determines the minimum number of required ESX hosts.
Space efficiency (vSAN HCI only)	Select the check box to enable vSAN deduplication and compression.
Mount Remote Datastore (vSAN Compute Cluster only)	Click Next and choose the remote datastore to provide storage for the new cluster.

vSAN ESA:

Option	Description
vSAN Cluster Type	• vSAN HCI: Provides storage and compute resources. • vSAN Storage: Provides storage resources only. You can mount a vSAN Storage datastore on other vSAN ESA or vSAN compute clusters.
Network Option (vSAN Storage only)	Select the vSAN Storage Client Network checkbox to isolate client or application network traffic from vSAN Storage internal traffic by utilizing dedicated VMkernel ports backed by physical network ports.
Storage Type	Local vSAN datastore. This setting cannot be modified.
Storage Policy	Auto-policy management. This setting cannot be modified. Based on the type of cluster and number of hosts, vSAN creates and assigns a default datastore policy for best capacity utilization after the cluster configuration is completed. Policy details can be viewed in the vSphere Client (Policies and Profiles > VM Storage Policies).

8. Enter the NFS Storage details and click **Next**.

This screen only displays if you selected NFS storage.

Option	Description
Datastore Name	Enter a name for the NFS datastore. The datastore name must contain 3-80 characters and cannot contain any leading or trailing whitespace. If you using a previously mounted NFS share, the datastore will be automatically renamed to original datastore name.
Folder	Enter the path to the NFS share. The folder location must contain 3-255 characters and the path cannot contain any whitespace.
NFS Server IP Address	Enter the IP address of the NFS server.

9. Enter the FC Storage details and click **Next**.

This screen only displays if you selected FC storage.

Option	Description
Datastore Name	Enter the name of the VMFS on FC datastore. Datastore name must contain 3-80 characters and cannot contain any leading or trailing whitespace.

10. Enter the vVol Storage details and click **Next**.

This screen only displays if you selected vVols storage.

Option	Description
VASA Protocol Type	Select a VASA protocol type. vVols supports FC, NFS, and iSCSI storage protocol types.
VASA Provider	Select a VASA provider name. The VASA Providers list is determined by the selected VASA protocol type.
Storage Container	Select a storage container. The Storage Containers list is determined by the selected VASA Provider. The list does not display storage containers that are already in use by other clusters.
VASA User	Select a VASA user. The VASA User list is determined by the selected VASA Provider.
Datastore Name	Enter a datastore name. The datastore name must contain 3-80 characters and cannot contain any leading or trailing whitespace.

11. On the Host Selection page, select hosts for the SDDC cluster and click **Next**.

You can use the toggle button to turn **Skip failed hosts during cluster creation** off or on. When this option is off, cluster creation will fail if you select an unhealthy host. When this option is on, cluster creation will succeed if you selected enough healthy hosts to meet the minimum requirements for a new cluster.

The hosts must be commissioned with the same storage type that you selected for the cluster.

When you have selected the minimum number of hosts required for this cluster, the **Next** button is enabled.

12. Enter the Distributed Switch details and click **Next**.

Option	Description
Default	This profile is recommended. It provides a unified fabric for all traffic types using a single vSphere Distributed Switch.
Storage Traffic Separation	This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for storage traffic, while the other is used for all other traffic.
NSX Traffic Separation	This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for NSX Edge and overlay traffic, while the other is used for all other traffic.
Storage Traffic and NSX Traffic Separation	This profile creates three vSphere Distributed Switches with separate physical NICs. One switch is dedicated for NSX Edge and overlay traffic, the second switch is dedicated for storage traffic, while the third is used for all other traffic.
Custom Switch Configuration	Copy from a preconfigured profile or create a new distributed switch configuration. Multiple vSphere Distributed Switches can be configured. Each vSphere Distributed Switch can hold one or more network traffic configurations. Some network traffic types are mandatory. Switch configuration is incomplete until these mandatory traffic types are configured. Management, vMotion, vSAN and NSX network types can be configured only once for a cluster. NSX and Public network traffic types can be configured in multiple vSphere Distributed Switches.

13. On the Review page, review the vSphere cluster details and click **Finish**.

To add another cluster in parallel, repeat the above steps.

Shrink a Workload Domain

You can reduce the management domain or a workload domain by removing a host from a vSphere cluster in the workload domain or by deleting a vSphere cluster.

Remove a Host from a vSphere Cluster in a Workload Domain

You can remove a host from a vSphere cluster in the management domain or a VI workload domain through the Workload Domains page in SDDC Manager UI.

- Delete any workload virtual machines created outside VMware Cloud Foundation or move them to another host.
- You cannot remove a host from a vSphere cluster if that host is hosting an NSX Edge node.
 - Move the NSX Edge node to another host in the same vSphere cluster. You cannot move an NSX Edge node to a host that already hosts another NSX Edge node.
 - If you cannot move the NSX Edge node, delete the NSX Edge node on the host. You cannot delete NSX Edge nodes if doing so would result in an NSX Edge cluster with fewer than two NSX Edge nodes.
- You cannot remove a host from a vSphere cluster if that vSphere cluster was selected for NSX Edge node placement and the NSX Edge node deployment is still in pending state.

Before you remove a host from a vSphere cluster, ensure that you have enough hosts remaining to facilitate the configured vSAN availability. Failure to do so might result in the datastore being marked as read-only or in data loss.

Note: You can add hosts to or remove hosts from multiple different vSphere clusters in parallel. For example, you remove three hosts from Cluster A, and while that task is running, you can start a separate task to remove (or add) four hosts to

Cluster B. See [VMware Configuration Maximums](#) for information about the maximum number of add/remove hosts tasks that you can run in parallel.

1. In the navigation pane, click **Inventory > Workload Domains**.
2. In the workload domains table, click the name of the workload domain that you want to modify.
3. Click the **Clusters** tab.
4. Click the name of the cluster from which you want to remove a host.
5. Click the **Hosts** tab.
6. Select the host to remove and click **Remove Selected Hosts**.
An alert appears, asking you to confirm or cancel the action. If the removal results in the number of hosts in the vSphere cluster being less than the minimum number of required hosts, you must click **Force Remove** to remove the host.
7. Click **Remove** to confirm the action.
The details page for the cluster appears with a message indicating that the host is being removed. When the removal process is complete, the host is removed from the hosts table.

Decommission the host. See [Decommission ESX Hosts](#). After you decommission a host, you must re-image it and commission it before you can use it again.

Delete a vSphere Cluster from a Workload Domain

You can delete a vSphere cluster from the management domain or from a workload domain. vSAN datastores on the ESXi hosts in the deleted cluster are destroyed, while other types of storage are unmounted.

- If remote vSAN datastores are mounted on the cluster, the cluster cannot be deleted. To delete such clusters, you must first migrate any VMs from the remote datastore to the local datastore and then unmount the vSAN remote datastores from vCenter.
- Migrate or back up the VMs and data on the datastore associated with the cluster to another location if you want the ability to restore them later.
- Delete the NSX Edge clusters hosted on the vSphere cluster or shrink the NSX Edge cluster by deleting NSX Edge nodes hosted on the vSphere cluster. You cannot delete NSX Edge nodes if doing so would result in an NSX Edge cluster with fewer than two NSX Edge nodes. For information about deleting an NSX Edge cluster, see [KB 78635](#).

You cannot delete the last cluster in a workload domain. Instead, delete the workload domain.

1. Log in to SDDC Manager at `https://<sddc_manager_fqdn>` with a user assigned the **Administrator** role.
2. In the navigation pane, click **Inventory > Workload Domains**.
3. Click the name of the workload domain that contains the vSphere cluster you want to delete.
4. Click the **Clusters** tab.
5. Click the vertical ellipsis (three dots) next to the cluster name and click **Delete Cluster**.



6. Click **Delete Cluster** to confirm that you want to delete the cluster.
You cannot perform additional workload domain tasks until the Delete Cluster workflow has completed.

Manage VCF Domain Configuration Drift

If you make out-of-band changes from vCenter or NSX Manager, you can use VCF Operations to update SDDC Manager with these changes and sync the SDDC Manager and vCenter inventories.

If the SDDC Manager and vCenter inventories get out of sync, some workflows may be blocked. You can use this procedure to unblock these tasks.

The sync workflow requires you to select a VCF domain to sync. If you must sync multiple workload domains, you run the sync workflow multiple times.

1. In VCF Operations, select **Inventory > Detailed View**.
2. Expand **VCF Instances** and browse to the VCF instance that contains the VCF domain you want to sync.
3. Click the VCF domain name.
4. Click **Actions > Sync Inventory**.
5. Review the information and click **Sync Inventory**.

If your existing NSX Manager instance has one or more Edge clusters, the sync process adds the discovered Edge node VMs to the VCF inventory, including their user credentials. During the sync process, the Edge node credential passwords will be reset. You can retrieve the updated passwords from the VCF credential store.

Select the **Enable Edge cluster sync and import NSX Edge node VMs** check box to perform the password reset operation for all the nodes in the existing Edge cluster, as well as for any nodes added later.

Stretching vSAN Clusters in VMware Cloud Foundation

You can stretch a vSAN cluster in a workload domain across two availability zones within a region. Both availability zones must contain an equal number of hosts to ensure failover in case any of the availability zones goes down.

VMware Cloud Foundation 9.0 supports stretching the following types of vSAN clusters using the SDDC Manager API:

- vSAN ESA/OSA
- vSAN compute
- vSAN storage (formerly known as vSAN Max)

In addition to stretching existing clusters, you can use the SDDC Manager API to create a stretched vSAN compute cluster. The following procedures provide the prerequisites and steps for stretching existing clusters, as well as creating a stretched vSAN compute cluster. You can use the examples as the basis for performing these tasks in your own environment. For more information about all the available parameters see the SDDC Manager API Reference Guide.

The default management domain vSphere cluster must be stretched before a workload domain cluster can be stretched. This ensures that the NSX control plane and management VMs (vCenter, NSX management cluster, and SDDC Manager) remain accessible if the stretched cluster in the primary availability zone goes down.

Note: You cannot stretch a cluster in the following conditions:

- The cluster shares a vSAN Storage Policy with any other clusters.
- The cluster includes DPU-backed hosts.
- The cluster includes hosts from different subnets or networks (L3).

Some use cases for stretching a cluster are described below.

- **Planned maintenance**
You can perform a planned maintenance on an availability zone without any downtime and then migrate the applications after the maintenance is completed.

- Automated recovery
Stretching a cluster automatically initiates VM restart and recovery, and has a low recovery time for the majority of unplanned failures.
- Disaster avoidance
With a stretched cluster, you can prevent service outages before an impending disaster.

For design details about stretched clusters, see [vSphere Stretched Cluster Model](#).

Deploy and Configure vSAN Witness Host

vSAN HCI and vSAN storage stretched clusters require a witness host deployed in a vSAN witness zone, which must be different from the location of both availability zones. vSAN compute stretched cluster do not require a witness host.

You deploy the vSAN witness host using an appliance instead of using a dedicated physical ESX host as a witness host. The witness host does not run virtual machines and must run the same version of ESX as the ESX hosts in the stretched cluster. It must also meet latency and Round Trip Time (RTT) requirements.

There are separate vSAN witness appliances for vSAN OSA and vSAN ESA. You must deploy the witness appliance that matches the cluster type that you are stretching.

Cluster Type	vSAN Witness Appliance
vSAN ESA	vSAN ESA appliance
vSAN OSA	vSAN OSA appliance
vSAN storage	vSAN ESA appliance
vSAN compute	n/a

Deploy vSAN Witness Host

You deploy the vSAN witness host for a stretched cluster at a site which is isolated from the existing availability zones to prevent propagation of failure or outage in the data center.

Download the VMware vSAN Witness Appliance .ova file from the [Broadcom Support Portal](#).

- For stretching a vSAN OSA cluster, download the appliance for vSAN OSA.
- For stretching a vSAN ESA cluster, download the appliance for vSAN ESA.
- For stretching a vSAN storage cluster, download the appliance for vSAN ESA.
- For stretching a vSAN compute cluster or creating a vSAN compute stretch cluster, no vSAN witness appliance is required.

- In a web browser, log in to vCenter at `https://vcenter_fqdn/ui`.
- Select **Menu > Hosts and Clusters**.
- In the inventory panel, expand **vCenter > Datacenter**.
- Right-click the cluster and select **Deploy OVF template**.
- On the **Select an OVF template** page, select **Local file**, click **Upload files**, browse to the location of the vSAN witness host OVA file, and click **Next**.
- On the **Select a name and folder** page, enter a name for the virtual machine and click **Next**.
- On the **Select a compute resource** page, select a compute resource, and click **Next**.
- On the **Review details** page, review the settings and click **Next**.
- On the **License agreements** page, accept the license agreement and click **Next**.
- On the **Configuration** page, select **Medium** and click **Next**.
- On the **Select storage** page, select a datastore and click **Next**.
- On the **Select networks** page, select a portgroup for the witness and management network, and click **Next**.
- On the **Customize template** page, enter the root password for the witness, select a network for vSAN Traffic, enter the details for each source network, and click **Next**.

If you do not specify management and secondary network details, DHCP is used.
- On the **Ready to complete** page, click **Finish** and wait for the process to complete.
- Power on the vSAN witness host.
 - In the inventory panel, navigate to **vCenter > Datacenter > Cluster**.
 - Right-click the vSAN witness VM and from the **Actions** menu, select **Power > Power on**.

Register vSAN Witness Host

Before you can configure the vSAN witness host, you must register it with vCenter.

- Use the vSphere Client to log in to the vCenter containing the cluster that you want to stretch.
- In the vSphere Client, navigate to the datacenter.
- Right-click the datacenter and select **Add Host**.

Important: You must add the vSAN witness host to the datacenter. Do not add it to a folder.
- On the **Name and location** page, enter the Fully Qualified Domain Name (FQDN) of the vSAN witness host and click **Next**.

Note: Do not use the IP address.
- On the **Connection settings** page, enter the root credentials for the witness host and click **Next**.
- On the **Host summary** page, review the summary of the host details and click **Next**.
- On the **Host lifecycle** page, select an option and specify a vSphere Lifecycle Manager image.
- On the **Assign license** page, assign an existing license and click **Next**.

Note: Do not create a new license.

- Review the summary and click **Finish**.

Configure NTP on the Witness Host

To prevent time synchronization issues, configure the NTP service on the vSAN witness host.

- In the inventory panel of the vSphere Client, select **vCenter > Datacenter**.
- Select the vSAN witness host and click the **Configure** tab.
- Configure the NTP client on the vSAN witness host.
 - In the **System** section, click **Time configuration** and click the **Edit** button.
 - Select **Use Network Time Protocol (enable NTP client)**.
 - Configure the following settings and click **OK**.

Setting	Value
NTP Servers	NTP server address
Start NTP Service	Selected
NTP Service Startup Policy	Start and stop with host

Configure the VMkernel Adapters on the vSAN Witness Host

To enable vSAN data network communication between the availability zones, configure the witness network on the vSAN witness host.

- In the inventory panel of the vSphere Client, select **vCenter > Datacenter**.
- Select the vSAN witness host and click the **Configure** tab.
- Remove the dedicated witness traffic VMkernel adapter on the vSAN Witness host.
 - In the **Networking** section, click **VMkernel adapters**.
 - Select the kernel adapter **vmk1** with **secondaryPg** as **Network label** and click **Remove**.
 - On the **Remove VMkernel adapter** dialog box, click **Remove**.
- Remove the secondary switch from the vSAN witness host.
 - In the left pane, select **Networking > Virtual switches**.
 - Expand the **Standard switch: secondary switch** section.
 - Click the vertical ellipsis and from the drop-down menu, select **Remove**.
 - On the **Remove standard switch** dialog box, click **Yes**.
- Remove the virtual machine network port group on the vSAN witness host.
 - Expand the **Standard switch: vSwitch0** section.
 - In the **VM Network** pane, click the vertical ellipsis and from the drop-down menu, select **Remove**.
 - On the **Remove port group** dialog box, click **Yes**.
- Enable witness traffic on the VMkernel adapter for the management network of the vSAN witness host.
 - On the **VMkernel adapters** page, select the **vmk0** adapter and click **Edit**.
 - In the **vmk0 - edit settings** dialog box, click **Port properties**, select the **vSAN** check box, and click **OK**.

Stretch a vSAN ESA or OSA Cluster Using the SDDC Manager API

You can stretch a vSAN cluster (ESA or OSA) in the management domain or workload domain using a JSON specification and the SDDC Manager API.

- See [vSAN Stretched HCI Cluster Storage Model](#) for design requirements and recommendations.

- Verify that vCenter is operational.
- Verify that you have completed the Planning and Preparation Workbook with the management domain or workload domain deployment option included.
- Verify that your environment meets the requirements listed in the Prerequisite Checklist sheet in the Planning and Preparation Workbook.
- Create a network pool for availability zone 2.
- Commission vSAN ESA or OSA hosts for availability zone 2. See [Commission ESX Hosts](#).
- Ensure that you have enough hosts such that there is an equal number of hosts on each availability zone. This is to ensure that there are sufficient resources in case an availability zone goes down completely.
- Deploy and configure a vSAN witness host. See [Deploy and Configure vSAN Witness Host](#).
- If you are stretching a cluster in a workload domain, the default management vSphere cluster must already be stretched.

Note: You cannot stretch a cluster in the following cases:

- The cluster shares a vSAN Storage Policy with any other clusters.
- The cluster includes DPU-backed hosts.
- The cluster includes hosts within an availability zone that have different subnets or networks (L3).
- The cluster has vSphere Supervisor enabled.

When you stretch a cluster, SDDC Manager modifies the site disaster tolerance setting for storage policy associated with datastore of that cluster from **None - standard cluster** to **Site mirroring - stretched cluster**. This affects all VMs using default datastore policy in that cluster. If you do not want to change the site disaster tolerance setting for specific VMs, apply a different storage policy to those VMs before stretching the cluster.

- Create a JSON specification file in a text editor.

The following example is for an environment with a single vSphere Distributed Switch. If you have multiple vSphere Distributed Switches, see the [SDDC Manager API Reference Guide](#) for details about creating a JSON specification.

Note: The ESX hosts that you are adding to availability zone 2 must use the same vmnic to vSphere Distributed Switch mapping as the existing hosts in availability zone 1.

```
{
  "clusterStretchSpec": {
    "deployWithoutLicenseKeys": true,
    "hostSpecs": [
      {
        "id": "<ESX host 1 ID>",
        "hostname": "sfo02-m01-r01-esx01.sfo.rainpole.io",
        "hostNetworkSpec": {
          "networkProfileName": "sfo02-m01-r01-network-profile01",
          "vmNics": [
            {
              "id": "vmnic0",
              "vdsName": "sfo-m01-cl01-vds01",
              "uplink": "uplink1"
            },
            {
              "id": "vmnic1",
              "vdsName": "sfo-m01-cl01-vds01",
              "uplink": "uplink2"
            }
          ]
        }
      }
    ]
  }
}
```

```

    },
    {
      "id": "<ESX host 2 ID>",
      "hostname": "sfo02-m01-r01-esx02.sfo.rainpole.io",
      "hostNetworkSpec": {
        "networkProfileName": "sfo02-m01-r01-network-profile01",
        "vmNics": [
          {
            "id": "vmnic0",
            "vdsName": "sfo-m01-cl01-vds01",
            "uplink": "uplink1"
          },
          {
            "id": "vmnic1",
            "vdsName": "sfo-m01-cl01-vds01",
            "uplink": "uplink2"
          }
        ]
      }
    },
    {
      "id": "<ESX host 3 ID>",
      "hostname": "sfo02-m01-r01-esx03.sfo.rainpole.io",
      "hostNetworkSpec": {
        "networkProfileName": "sfo02-m01-r01-network-profile01",
        "vmNics": [
          {
            "id": "vmnic0",
            "vdsName": "sfo-m01-cl01-vds01",
            "uplink": "uplink1"
          },
          {
            "id": "vmnic1",
            "vdsName": "sfo-m01-cl01-vds01",
            "uplink": "uplink2"
          }
        ]
      }
    },
    {
      "id": "<ESX host 4 ID>",
      "hostname": "sfo02-m01-r01-esx04.sfo.rainpole.io",
      "hostNetworkSpec": {
        "networkProfileName": "sfo02-m01-r01-network-profile01",
        "vmNics": [
          {
            "id": "vmnic0",
            "vdsName": "sfo-m01-cl01-vds01",
            "uplink": "uplink1"
          },
          {
            "id": "vmnic1",
            "vdsName": "sfo-m01-cl01-vds01",

```

```

      "uplink": "uplink2"
    }
  ]
},
"networkSpec": {
  "networkProfiles": [
    {
      "isDefault": true,
      "name": "sfo02-m01-r01-network-profile01",
      "nsxtHostSwitchConfigs": [
        {
          "ipAddressPoolName": "sfo02-m01-r01-ip-pool01-host",
          "uplinkProfileName": "sfo02-m01-r01-uplink-profile01",
          "vdsName": "sfo-m01-cl01-vds01",
          "vdsUplinkToNsxUplink": [
            {
              "nsxUplinkName": "uplink1",
              "vdsUplinkName": "uplink1"
            },
            {
              "nsxUplinkName": "uplink2",
              "vdsUplinkName": "uplink2"
            }
          ]
        }
      ]
    }
  ],
  "nsxClusterSpec": {
    "ipAddressPoolsSpec": [
      {
        "name": "sfo02-m01-r01-ip-pool01-host",
        "subnets": [
          {
            "cidr": "10.12.14.0/24",
            "gateway": "10.12.14.1",
            "ipAddressPoolRanges": [
              {
                "start": "10.12.14.101",
                "end": "10.12.14.132"
              }
            ]
          }
        ]
      }
    ]
  },
  "uplinkProfiles": [
    {
      "name": "sfo02-m01-r01-uplink-profile01",
      "transportVlan": 1214,
      "teamings": [

```

```

    {
      "name": "DEFAULT",
      "policy": "LOADBALANCE_SRCID",
      "standByUplinks": [],
      "activeUplinks": [
        "uplink1",
        "uplink2"
      ]
    }
  ]
}
}
}
},
"isEdgeClusterConfiguredForMultiAZ": true,
"witnessSpec": {
  "fqdn": "sfo-m01-cl01-vsw01.sfo.rainpole.io",
  "vsanCidr": "10.21.10.0/24",
  "vsanIp": "10.21.10.218"
},
"witnessTrafficSharedWithVsanTraffic": false
}
}

```

Note:

Replace the example values in the JSON file with the correct values for your environment.

`isEdgeClusterConfiguredForMultiAZ` should be set to `"true"` if the cluster hosts an NSX Edge cluster and `"false"` if the cluster does not host an NSX Edge cluster.

2. Log in to VCF Operations.
3. In the navigation pane, click **Developer Center > APIs & SDKs**.
4. Click the **API Explorer** link on the SDDC Manager API.

Click the VCF Instance which contains the cluster you want to stretch.
5. Retrieve and replace the unique IDs for each ESX host in the JSON specification file.
 - a) Navigate to **Hosts > GET /v1/hosts**.
 - b) In the **Status** text box, enter `UNASSIGNED_USEABLE` and click **Execute**.
 - c) In the **Response** section, click **PageOfHost**, copy the `id` element of each host, and replace the respective value in the JSON specification file.

ESX Host	Value
ESX Host 1	<i>ESX host 1 ID</i>
ESX Host 2	<i>ESX host 2 ID</i>
ESX Host 3	<i>ESX host 3 ID</i>
ESX Host 4	<i>ESX host 4 ID</i>

6. Retrieve the unique ID for the cluster you want to stretch.
 - a) Navigate to **Clusters > GET /v1/clusters**.
 - b) Click **Execute**.
 - c) In the **Response** section, click **PageOfCluster**, copy the `id` element of the cluster.

You will need the cluster ID for subsequent steps.
7. Validate the JSON specification file.
 - a) Navigate to **Clusters > POST /v1/clusters/{id}/validations**.
 - b) In the **Value** text box, enter the unique ID for the cluster that you retrieved in step 5.
 - c) In the **clusterUpdateSpec** text box, type

```

{
  "clusterUpdateSpec":
}

```

- d) Paste the JSON specification.

For example:

```

{
  "clusterUpdateSpec": {
    "clusterStretchSpec": {
      "hostSpecs": [
        {
          "hostname": "sfo02-w01-esx01.sfo.rainpole.io",
          "hostNetworkSpec": {
            "networkProfileName": "sfo-w01-az2-nsx-np01",
            "vmNics": [
              {
                "id": "vmnic0",
                "uplink": "uplink1",
                "vdsName": "sfo-w01-cl01-vds01"
              },
              {
                "id": "vmnic1",
                "uplink": "uplink2",
                "vdsName": "sfo-w01-cl01-vds01"
              }
            ],
            "id": "<ESX host 1 ID>"
          },
          {
            "hostname": "sfo02-w01-esx02.sfo.rainpole.io",
            "hostNetworkSpec": {
              "networkProfileName": "sfo-w01-az2-nsx-np01",
              "vmNics": [
                {
                  "id": "vmnic0",
                  "uplink": "uplink1",
                  "vdsName": "sfo-w01-cl01-vds01"
                },
                {
                  "id": "vmnic1",
                  "uplink": "uplink2",

```

```

    "vdsName": "sfo-w01-cl01-vds01"
  }
},
"id": "<ESX host 2 ID>"
},
{
  "hostname": "sfo02-w01-esx03.sfo.rainpole.io",
  "hostNetworkSpec": {
    "networkProfileName": "sfo-w01-az2-nsx-np01",
    "vmNics": [
      {
        "id": "vmnic0",
        "uplink": "uplink1",
        "vdsName": "sfo-w01-cl01-vds01"
      },
      {
        "id": "vmnic1",
        "uplink": "uplink2",
        "vdsName": "sfo-w01-cl01-vds01"
      }
    ]
  },
  "id": "<ESX host 3 ID>"
},
{
  "hostname": "sfo02-w01-esx04.sfo.rainpole.io",
  "hostNetworkSpec": {
    "networkProfileName": "sfo-w01-az2-nsx-np01",
    "vmNics": [
      {
        "id": "vmnic0",
        "uplink": "uplink1",
        "vdsName": "sfo-w01-cl01-vds01"
      },
      {
        "id": "vmnic1",
        "uplink": "uplink2",
        "vdsName": "sfo-w01-cl01-vds01"
      }
    ]
  },
  "id": "<ESX host 4 ID>"
},
{
  "isEdgeClusterConfiguredForMultiAZ": "true",
  "networkSpec": {
    "networkProfiles": [
      {
        "isDefault": false,
        "name": "sfo-w01-az2-nsx-np01",
        "nsxtHostSwitchConfigs": [

```

```

    "ipAddressPoolName": "sfo-w01-az2-host-ip-pool01",
    "uplinkProfileName": "sfo-w01-az2-host-uplink-profile01",
    "vdsName": "sfo-w01-cl01-vds01",
    "vdsUplinkToNsxUplink": [
      {
        "nsxUplinkName": "uplink-1",
        "vdsUplinkName": "uplink1"
      },
      {
        "nsxUplinkName": "uplink-2",
        "vdsUplinkName": "uplink2"
      }
    ]
  },
  "nsxClusterSpec": {
    "ipAddressPoolsSpec": [
      {
        "description": "WLD01 AZ2 Host TEP Pool",
        "name": "sfo-w01-az2-host-ip-pool01",
        "subnets": [
          {
            "cidr": "172.16.44.0/24",
            "gateway": "172.16.44.253",
            "ipAddressPoolRanges": [
              {
                "end": "172.16.44.200",
                "start": "172.16.44.10"
              }
            ]
          }
        ]
      }
    ],
    "uplinkProfiles": [
      {
        "name": "sfo-w01-az2-host-uplink-profile01",
        "teamings": [
          {
            "activeUplinks": [
              "uplink-1",
              "uplink-2"
            ],
            "name": "DEFAULT",
            "policy": "LOADBALANCE_SRCID",
            "standByUplinks": []
          }
        ],
        "transportVlan": 1644
      }
    ]
  }

```

```

    }
  },
  "witnessSpec": {
    "fqdn": "sfo-w01-cl01-vsw01.sfo.rainpole.io",
    "vsanCidr": "172.17.11.0/24",
    "vsanIp": "172.17.11.219"
  },
  "witnessTrafficSharedWithVsanTraffic": false
}
}

```

Note:

Replace the example values in the JSON file with the correct values for your environment.

`isEdgeClusterConfiguredForMultiAZ` should be set to `"true"` if the cluster hosts an NSX Edge cluster and `"false"` if the cluster does not host an NSX Edge cluster.

- e) Click **Execute**.
 - f) In the confirmation dialog box, click **Continue**.
 - g) In the **Response** section, expand the **result** section and verify that the response is `SUCCEEDED`.
8. Stretch the cluster with the JSON specification.
- a) Navigate to **Clusters > PATCH /v1/clusters/{id}**.
 - b) Paste the unique ID of the cluster in the **Value** text-box.
 - c) In the **clusterUpdateSpec** text box, paste the JSON specification.

For example:

```

{
  "clusterStretchSpec": {
    "deployWithoutLicenseKeys": true,
    "hostSpecs": [
      {
        "hostname": "sfo02-w01-esx01.sfo.rainpole.io",
        "hostNetworkSpec": {
          "networkProfileName": "sfo-w01-az2-nsx-np01",
          "vmNics": [
            {
              "id": "vmnic0",
              "uplink": "uplink1",
              "vdsName": "sfo-w01-cl01-vds01"
            },
            {
              "id": "vmnic1",
              "uplink": "uplink2",
              "vdsName": "sfo-w01-cl01-vds01"
            }
          ],
          "id": "<ESX host 1 ID>"
        },
        {
          "hostname": "sfo02-w01-esx02.sfo.rainpole.io",
          "hostNetworkSpec": {
            "networkProfileName": "sfo-w01-az2-nsx-np01",

```

```

    "vmNics": [
      {
        "id": "vmnic0",
        "uplink": "uplink1",
        "vdsName": "sfo-w01-cl01-vds01"
      },
      {
        "id": "vmnic1",
        "uplink": "uplink2",
        "vdsName": "sfo-w01-cl01-vds01"
      }
    ],
    "id": "<ESX host 2 ID>"
  },
  {
    "hostname": "sfo02-w01-esx03.sfo.rainpole.io",
    "hostNetworkSpec": {
      "networkProfileName": "sfo-w01-az2-nsx-np01",
      "vmNics": [
        {
          "id": "vmnic0",
          "uplink": "uplink1",
          "vdsName": "sfo-w01-cl01-vds01"
        },
        {
          "id": "vmnic1",
          "uplink": "uplink2",
          "vdsName": "sfo-w01-cl01-vds01"
        }
      ],
      "id": "<ESX host 3 ID>"
    },
    {
      "hostname": "sfo02-w01-esx04.sfo.rainpole.io",
      "hostNetworkSpec": {
        "networkProfileName": "sfo-w01-az2-nsx-np01",
        "vmNics": [
          {
            "id": "vmnic0",
            "uplink": "uplink1",
            "vdsName": "sfo-w01-cl01-vds01"
          },
          {
            "id": "vmnic1",
            "uplink": "uplink2",
            "vdsName": "sfo-w01-cl01-vds01"
          }
        ],
        "id": "<ESX host 4 ID>"
      }
    }
  ]
}

```

```

    ],
    "isEdgeClusterConfiguredForMultiAZ": "true",
    "networkSpec": {
      "networkProfiles": [
        {
          "isDefault": false,
          "name": "sfo-w01-az2-nsx-np01",
          "nsxtHostSwitchConfigs": [
            {
              "ipAddressPoolName": "sfo-w01-az2-host-ip-pool01",
              "uplinkProfileName": "sfo-w01-az2-host-uplink-profile01",
              "vdsName": "sfo-w01-cl01-vds01",
              "vdsUplinkToNsxUplink": [
                {
                  "nsxUplinkName": "uplink-1",
                  "vdsUplinkName": "uplink1"
                },
                {
                  "nsxUplinkName": "uplink-2",
                  "vdsUplinkName": "uplink2"
                }
              ]
            }
          ]
        }
      ],
      "nsxClusterSpec": {
        "ipAddressPoolsSpec": [
          {
            "description": "WLD01 AZ2 Host TEP Pool",
            "name": "sfo-w01-az2-host-ip-pool01",
            "subnets": [
              {
                "cidr": "172.16.44.0/24",
                "gateway": "172.16.44.253",
                "ipAddressPoolRanges": [
                  {
                    "end": "172.16.44.200",
                    "start": "172.16.44.10"
                  }
                ]
              }
            ]
          }
        ],
        "uplinkProfiles": [
          {
            "name": "sfo-w01-az2-host-uplink-profile01",
            "teamings": [
              {
                "activeUplinks": [
                  "uplink-1",
                  "uplink-2"
                ]
              }
            ]
          }
        ]
      }
    }
  ],
  "nsxUplinkToNsxUplink": [
    {
      "nsxUplinkName": "uplink-1",
      "vdsUplinkName": "uplink1"
    },
    {
      "nsxUplinkName": "uplink-2",
      "vdsUplinkName": "uplink2"
    }
  ]
}

```

```

    ],
    "name": "DEFAULT",
    "policy": "LOADBALANCE_SRCID",
    "standByUplinks": []
  }
},
"transportVlan": 1644
}
]
},
"witnessSpec": {
  "fqdn": "sfo-w01-cl01-vsw01.sfo.rainpole.io",
  "vsanCidr": "172.17.11.0/24",
  "vsanIp": "172.17.11.219"
},
"witnessTrafficSharedWithVsanTraffic": false
}
}

```

Note:

Replace the example values in the JSON file with the correct values for your environment.

`isEdgeClusterConfiguredForMultiAZ` should be set to "true" if the cluster hosts an NSX Edge cluster and "false" if the cluster does not host an NSX Edge cluster.

d) Click **Execute**.

e) On the confirmation dialog box, click **Continue**.

You can monitor the stretch cluster creation in **Fleet Management > Tasks**.

Create a vSAN Compute Stretch Cluster Using the SDDC Manager API

You can create a vSAN compute stretch cluster using the POST /v1/clusters API.

- See [vSAN Stretched Compute Cluster Model](#) for design requirements and recommendations.
- A minimum of four ESX hosts commissioned with the vSAN type vSAN Compute Cluster. See [Commission ESX Hosts](#).
- ESX, vCenter, and SDDC Manager must all be version 9.0 or later.
- An existing remote datastore. It can be a vSAN ESA, vSAN OSA, or vSAN storage datastore.
- The server cluster providing the remote datastore must be stretched. See [Stretch a vSAN Storage Cluster Using the SDDC Manager API](#).

No vSAN witness host is required for a stretched vSAN compute cluster.

If you want to stretch a vSAN compute cluster that you already created, see [Stretch a vSAN Compute Cluster Using the SDDC Manager API](#).

This procedure provides two different sample JSON specification files for the different topologies:

Topology	Description
Symmetric	This reflects a topology that has multiple routes to and from the server cluster, with all routes having the same bandwidth and latency characteristics. This is not typical in stretched cluster deployments but may arise if the vSAN compute cluster is deployed at different sites unrelated to where the stretched vSAN storage cluster resides.

Topology	Description
Asymmetric	This typically reflects a topology that has multiple routes to and from the server cluster, with one route having optimal bandwidth and latency characteristics. This is typical in stretched cluster deployments where a client stretched cluster needs to access storage on a server stretched cluster would ideally access the object data in the same site, known as inter-site-connectivity. In this case you specify the site affinity between server and client sites in the JSON specification file. For example, if your environment looks like this: - AZ1 (Server): East Coast - AZ2 (Server): West Coast - AZ1 (Client): East Coast - AZ2 (Client): West Coast Then you would use an asymmetric topology with the following site affinity: - AZ1 (Client) → AZ1 (Server) - AZ2 (Client) → AZ2 (Server)

1. Create a JSON specification file in a text editor.

Symmetric vSAN remote datastore topology:

```
{
  "domainId": "<domain ID>",
  "computeSpec": {
    "clusterSpecs": [
      {
        "name": "sfo-w01-cl02",
        "hostSpecs": [
          {
            "id": "ESX host 1 ID",
            "hostname": "sfo01-w01-r01-esx04.sfo.rainpole.io",
            "azName": "sfo-w01-r01-c01_primary-az-faultdomain",
            "hostNetworkSpec": {
              "networkProfileName": "sfo01-w01-r01-network-profile01",
              "vmNics": [
                {
                  "id": "vmnic0",
                  "vdsName": "sfo-w01-cl02-vds01",
                  "uplink": "uplink1"
                },
                {
                  "id": "vmnic1",
                  "vdsName": "sfo-w01-cl02-vds01",
                  "uplink": "uplink2"
                }
              ]
            }
          }
        ]
      },
      {
        "id": "ESX host 2 ID",
```

```
    "hostname": "sfo01-w01-r01-esx05.sfo.rainpole.io",
    "azName": "sfo-w01-r01-c01_primary-az-faultdomain",
    "hostNetworkSpec": {
      "networkProfileName": "sfo01-w01-r01-network-profile01",
      "vmNics": [
        {
          "id": "vmnic0",
          "vdsName": "sfo-w01-cl02-vds01",
          "uplink": "uplink1"
        },
        {
          "id": "vmnic1",
          "vdsName": "sfo-w01-cl02-vds01",
          "uplink": "uplink2"
        }
      ]
    }
  },
  {
    "id": "ESX host 3 ID",
    "hostname": "sfo02-w01-r01-esx04.sfo.rainpole.io",
    "azName": "sfo-w01-r01-c01_secondary-az-faultdomain",
    "hostNetworkSpec": {
      "networkProfileName": "sfo02-w01-r01-network-profile01",
      "vmNics": [
        {
          "id": "vmnic0",
          "vdsName": "sfo-w01-cl02-vds01",
          "uplink": "uplink1"
        },
        {
          "id": "vmnic1",
          "vdsName": "sfo-w01-cl02-vds01",
          "uplink": "uplink2"
        }
      ]
    }
  },
  {
    "id": "ESX host 4 ID",
    "hostname": "sfo02-w01-r01-esx05.sfo.rainpole.io",
    "azName": "sfo-w01-r01-c01_secondary-az-faultdomain",
    "hostNetworkSpec": {
      "networkProfileName": "sfo02-w01-r01-network-profile01",
      "vmNics": [
        {
          "id": "vmnic0",
          "vdsName": "sfo-w01-cl02-vds01",
          "uplink": "uplink1"
        },
        {
          "id": "vmnic1",
          "vdsName": "sfo-w01-cl02-vds01",
```

```

        "uplink": "uplink2"
    }
}
},
"datastoreSpec": {
    "vsanRemoteDatastoreClusterSpec": {
        "vsanRemoteDatastoreSpec": [
            {
                "datastoreUuid": "<-- ENTER UUID OF REMOTE DATASTORE HERE -->",
                "networkTopology": "Symmetric"
            }
        ]
    }
},
"networkSpec": {
    "vdsSpecs": [
        {
            "name": "sfo-w01-cl02-vds01",
            "mtu": "9000",
            "portGroupSpecs": [
                {
                    "name": "sfo-w01-cl02-pg-esxi-mgmt",
                    "transportType": "MANAGEMENT",
                    "standByUplinks": [],
                    "teamingPolicy": "loadbalance_loadbased",
                    "activeUplinks": [
                        "uplink1",
                        "uplink2"
                    ]
                },
                {
                    "name": "sfo-w01-cl02-pg-vmotion",
                    "transportType": "VMOTION",
                    "standByUplinks": [],
                    "teamingPolicy": "loadbalance_loadbased",
                    "activeUplinks": [
                        "uplink1",
                        "uplink2"
                    ]
                },
                {
                    "name": "sfo-w01-cl02-pg-vsan",
                    "transportType": "VSAN",
                    "standByUplinks": [],
                    "teamingPolicy": "loadbalance_loadbased",
                    "activeUplinks": [
                        "uplink1",
                        "uplink2"
                    ]
                }
            ]
        }
    ]
},

```

```

    "nsxtSwitchConfig": {
        "hostSwitchOperationalMode": "STANDARD",
        "transportZones": [
            {
                "name": "vlan-tz-sfo-w01-nsx01",
                "transportType": "VLAN"
            },
            {
                "name": "overlay-tz-sfo-w01-nsx01",
                "transportType": "OVERLAY"
            }
        ]
    }
},
"nsxClusterSpec": {
    "nsxTClusterSpec": {
        "ipAddressPoolsSpec": [
            {
                "name": "sfo01-w01-r01-ip-pool01-host",
                "subnets": [
                    {
                        "cidr": "10.13.14.0/24",
                        "gateway": "10.13.14.1",
                        "ipAddressPoolRanges": [
                            {
                                "start": "10.13.14.101",
                                "end": "10.13.14.132"
                            }
                        ]
                    }
                ]
            }
        ],
        {
            "name": "sfo02-w01-r01-ip-pool01-host",
            "subnets": [
                {
                    "cidr": "10.14.14.0/24",
                    "gateway": "10.14.14.1",
                    "ipAddressPoolRanges": [
                        {
                            "start": "10.14.14.101",
                            "end": "10.14.14.132"
                        }
                    ]
                }
            ]
        }
    ],
    "uplinkProfiles": [
        {
            "name": "sfo01-w01-r01-uplink-profile01",
            "transportVlan": 1314,

```

```

    "teamings": [
      {
        "name": "DEFAULT",
        "policy": "LOADBALANCE_SRCID",
        "standByUplinks": [],
        "activeUplinks": [
          "uplink1",
          "uplink2"
        ]
      }
    ]
  },
  {
    "name": "sfo02-w01-r01-uplink-profile01",
    "transportVlan": 1414,
    "teamings": [
      {
        "name": "DEFAULT",
        "policy": "LOADBALANCE_SRCID",
        "standByUplinks": [],
        "activeUplinks": [
          "uplink1",
          "uplink2"
        ]
      }
    ]
  }
],
"networkProfiles": [
  {
    "isDefault": true,
    "name": "sfo01-w01-r01-network-profile01",
    "nsxtHostSwitchConfigs": [
      {
        "ipAddressPoolName": "sfo01-w01-r01-ip-pool01-host",
        "uplinkProfileName": "sfo01-w01-r01-uplink-profile01",
        "vdsName": "sfo-w01-cl02-vds01",
        "vdsUplinkToNsxUplink": [
          {
            "nsxUplinkName": "uplink1",
            "vdsUplinkName": "uplink1"
          },
          {
            "nsxUplinkName": "uplink2",
            "vdsUplinkName": "uplink2"
          }
        ]
      }
    ]
  }
],
{

```

```

    "isDefault": false,
    "name": "sfo02-w01-r01-network-profile01",
    "nsxtHostSwitchConfigs": [
      {
        "ipAddressPoolName": "sfo02-w01-r01-ip-pool01-host",
        "uplinkProfileName": "sfo02-w01-r01-uplink-profile01",
        "vdsName": "sfo-w01-cl02-vds01",
        "vdsUplinkToNsxUplink": [
          {
            "nsxUplinkName": "uplink1",
            "vdsUplinkName": "uplink1"
          },
          {
            "nsxUplinkName": "uplink2",
            "vdsUplinkName": "uplink2"
          }
        ]
      }
    ]
  },
  {
    "clusterImageId": "<personality ID>"
  }
],
"deployWithoutLicenseKeys": "true"
}

```

Asymmetric vSAN remote datastore topology:

```

{
  "domainId": "<domain ID>",
  "computeSpec": {
    "clusterSpecs": [
      {
        "name": "sfo-w01-cl02",
        "hostSpecs": [
          {
            "id": "ESX host 1 ID",
            "hostname": "sfo01-w01-r01-esx04.sfo.rainpole.io",
            "azName": "sfo-w01-r01-c01_primary-az-faultdomain",
            "hostNetworkSpec": {
              "networkProfileName": "sfo01-w01-r01-network-profile01",
              "vmNics": [
                {
                  "id": "vmnic0",
                  "vdsName": "sfo-w01-cl02-vds01",
                  "uplink": "uplink1"
                },
                {
                  "id": "vmnic1",
                  "vdsName": "sfo-w01-cl02-vds01",
                  "uplink": "uplink2"
                }
              ]
            }
          }
        ]
      }
    ]
  }
}

```

```

    }
  }
},
{
  "id": "ESX host 2 ID",
  "hostname": "sfo01-w01-r01-esx05.sfo.rainpole.io",
  "azName": "sfo-w01-r01-c01_primary-az-faultdomain",
  "hostNetworkSpec": {
    "networkProfileName": "sfo01-w01-r01-network-profile01",
    "vmNics": [
      {
        "id": "vmnic0",
        "vdsName": "sfo-w01-cl02-vds01",
        "uplink": "uplink1"
      },
      {
        "id": "vmnic1",
        "vdsName": "sfo-w01-cl02-vds01",
        "uplink": "uplink2"
      }
    ]
  }
},
{
  "id": "ESX host 3 ID",
  "hostname": "sfo02-w01-r01-esx04.sfo.rainpole.io",
  "azName": "sfo-w01-r01-c01_secondary-az-faultdomain",
  "hostNetworkSpec": {
    "networkProfileName": "sfo02-w01-r01-network-profile01",
    "vmNics": [
      {
        "id": "vmnic0",
        "vdsName": "sfo-w01-cl02-vds01",
        "uplink": "uplink1"
      },
      {
        "id": "vmnic1",
        "vdsName": "sfo-w01-cl02-vds01",
        "uplink": "uplink2"
      }
    ]
  }
},
{
  "id": "ESX host 4 ID",
  "hostname": "sfo02-w01-r01-esx05.sfo.rainpole.io",
  "azName": "sfo-w01-r01-c01_secondary-az-faultdomain",
  "hostNetworkSpec": {
    "networkProfileName": "sfo02-w01-r01-network-profile01",
    "vmNics": [
      {
        "id": "vmnic0",

```

```

    "vdsName": "sfo-w01-cl02-vds01",
    "uplink": "uplink1"
  },
  {
    "id": "vmnic1",
    "vdsName": "sfo-w01-cl02-vds01",
    "uplink": "uplink2"
  }
]
}
},
{
  "datastoreSpec": {
    "vsanRemoteDatastoreClusterSpec": {
      "vsanRemoteDatastoreSpec": [
        {
          "datastoreUuid": "<-- ENTER UUID OF REMOTE DATASTORE HERE -->",
          "networkTopology": "Asymmetric",
          "siteAffinity": [
            {
              "serverSite": "sfo-w01-c01_primary-az-faultdomain",
              "clientSite": "sfo-w01-c01_primary-az-faultdomain"
            },
            {
              "serverSite": "sfo-w01-c01_secondary-az-faultdomain",
              "clientSite": "sfo-w01-c01_secondary-az-faultdomain"
            }
          ]
        }
      ]
    }
  },
  "networkSpec": {
    "vdsSpecs": [
      {
        "name": "sfo-w01-cl02-vds01",
        "mtu": "9000",
        "portGroupSpecs": [
          {
            "name": "sfo-w01-cl02-pg-esxi-mgmt",
            "transportType": "MANAGEMENT",
            "standByUplinks": [],
            "teamingPolicy": "loadbalance_loadbased",
            "activeUplinks": [
              "uplink1",
              "uplink2"
            ]
          },
          {
            "name": "sfo-w01-cl02-pg-vmotion",
            "transportType": "VMOTION",
            "standByUplinks": [],
            "teamingPolicy": "loadbalance_loadbased",

```

```

        "activeUplinks": [
            "uplink1",
            "uplink2"
        ]
    },
    {
        "name": "sfo-w01-cl02-pg-vsan",
        "transportType": "VSAN",
        "standByUplinks": [],
        "teamingPolicy": "loadbalance_loadbased",
        "activeUplinks": [
            "uplink1",
            "uplink2"
        ]
    }
],
"nsxtSwitchConfig": {
    "hostSwitchOperationalMode": "STANDARD",
    "transportZones": [
        {
            "name": "vlan-tz-sfo-w01-nsx01",
            "transportType": "VLAN"
        },
        {
            "name": "overlay-tz-sfo-w01-nsx01",
            "transportType": "OVERLAY"
        }
    ]
}
},
"nsxClusterSpec": {
    "nsxTClusterSpec": {
        "ipAddressPoolsSpec": [
            {
                "name": "sfo01-w01-r01-ip-pool101-host",
                "subnets": [
                    {
                        "cidr": "10.13.14.0/24",
                        "gateway": "10.13.14.1",
                        "ipAddressPoolRanges": [
                            {
                                "start": "10.13.14.101",
                                "end": "10.13.14.132"
                            }
                        ]
                    }
                ]
            }
        ]
    }
},
{
    "name": "sfo02-w01-r01-ip-pool101-host",
    "subnets": [
        {

```

```

        "cidr": "10.14.14.0/24",
        "gateway": "10.14.14.1",
        "ipAddressPoolRanges": [
            {
                "start": "10.14.14.101",
                "end": "10.14.14.132"
            }
        ]
    }
}
],
"uplinkProfiles": [
    {
        "name": "sfo01-w01-r01-uplink-profile01",
        "transportVlan": 1314,
        "teamings": [
            {
                "name": "DEFAULT",
                "policy": "LOADBALANCE_SRCID",
                "standByUplinks": [],
                "activeUplinks": [
                    "uplink1",
                    "uplink2"
                ]
            }
        ]
    }
],
{
    "name": "sfo02-w01-r01-uplink-profile01",
    "transportVlan": 1414,
    "teamings": [
        {
            "name": "DEFAULT",
            "policy": "LOADBALANCE_SRCID",
            "standByUplinks": [],
            "activeUplinks": [
                "uplink1",
                "uplink2"
            ]
        }
    ]
}
],
"networkProfiles": [
    {
        "isDefault": true,
        "name": "sfo01-w01-r01-network-profile01",
        "nsxtHostSwitchConfigs": [
            {
                "ipAddressPoolName": "sfo01-w01-r01-ip-pool101-host",

```

```

    "uplinkProfileName": "sfo01-w01-r01-uplink-profile01",
    "vdsName": "sfo-w01-cl02-vds01",
    "vdsUplinkToNsxUplink": [
      {
        "nsxUplinkName": "uplink1",
        "vdsUplinkName": "uplink1"
      },
      {
        "nsxUplinkName": "uplink2",
        "vdsUplinkName": "uplink2"
      }
    ]
  },
  {
    "isDefault": false,
    "name": "sfo02-w01-r01-network-profile01",
    "nsxtHostSwitchConfigs": [
      {
        "ipAddressPoolName": "sfo02-w01-r01-ip-pool01-host",
        "uplinkProfileName": "sfo02-w01-r01-uplink-profile01",
        "vdsName": "sfo-w01-cl02-vds01",
        "vdsUplinkToNsxUplink": [
          {
            "nsxUplinkName": "uplink1",
            "vdsUplinkName": "uplink1"
          },
          {
            "nsxUplinkName": "uplink2",
            "vdsUplinkName": "uplink2"
          }
        ]
      }
    ]
  }
],
{
  "clusterImageId": "<personality ID>"
},
{
  "deployWithoutLicenseKeys": "true"
}

```

Note: Replace the example values in the JSON file with the correct values for your environment.

2. Log in to VCF Operations.
3. In the navigation pane, click **Developer Center > APIs & SDKs**.
4. Click the **API Explorer** link on the SDDC Manager API.
5. Click the VCF Instance in which you want to create the stretched cluster.
6. Retrieve the unique IDs for each ESX host that will be part of the stretched cluster and add the IDs to the JSON specification file.
 - a) Navigate to **Hosts > GET /v1/hosts**.
 - b) In the **Status** text box, enter `UNASSIGNED_USEABLE` and click **Execute**.
 - c) In the **Response** section, click **PageOfHost**, copy the `id` element of each host, and replace the respective value in the JSON specification file.

ESX Host	Value
ESX Host 1	<i>ESX host 1 ID</i>
ESX Host 2	<i>ESX host 2 ID</i>
ESX Host 3	<i>ESX host 3 ID</i>
ESX Host 4	<i>ESX host 4 ID</i>

7. Retrieve the ID of the workload domain in which you want to create the cluster and add the ID to the JSON specification file.
 - a) Navigate to **Domains > GET /v1/domains**.
 - b) Click **Execute**.
 - c) In the **Response** section, click the domain name, copy the `id` element, and replace the `<domain ID>` value in the JSON specification file.
8. Retrieve the vSphere Lifecycle Manager image ID to use for the cluster and add the ID to the JSON specification file.
 - a) Navigate to **Personalities > GET /v1/personalities**.
 - b) Click **Execute**.
 - c) In the **Response** section, click the personality representing the image you want to use, copy the `personalityId` element, and replace the `<personality ID>` value in the JSON specification file.
9. Retrieve the datastore Uuid for the server cluster providing the remote datastore and add it to the JSON specification file.
 - a) Navigate to **Clusters > GET /v1/clusters**.
 - b) Click **Execute**.
 - c) In the **Response** section, click **PageOfCluster**, and copy the `id` element of the server cluster providing the remote datastore.
 - d) Navigate to **Clusters > GET /v1/clusters{id}/datastores**.
 - e) In the **Value** text box, enter the unique ID for the server cluster and click **Execute**.
 - f) In the **Response** section, click the datastore, copy the `id` element, and replace the `<datastore ID>` value in the JSON specification file.
10. Navigate to **Clusters > POST /v1/clusters**.
11. Paste the content from your JSON specification file in the **Value** text box for the `body` parameter and click **Execute**.
12. On the confirmation dialog box, click **Continue**.

You can monitor the stretch cluster creation in **Fleet Management > Tasks**.

Stretch a vSAN Compute Cluster Using the SDDC Manager API

You can stretch an existing vSAN compute cluster using the PATCH /v1/clusters/{id} API.

- See [vSAN Stretched Compute Cluster Model](#) for design requirements and recommendations.
- A minimum of three ESX hosts commissioned with the vSAN type vSAN Compute Cluster. See [Commission ESX Hosts](#).
- An existing vSAN compute cluster with at least three ESX hosts.
- ESX, vCenter, and SDDC Manager must all be version 9.0 or later.
- The server cluster providing the remote datastore must be stretched. See [Stretch a vSAN Storage Cluster Using the SDDC Manager API](#).
- Unmount all datastores from the vSAN compute cluster, except for the primary datastore.
- Unregister all VMs running on the primary datastore.

No vSAN witness host is required for a stretched vSAN compute cluster.

If you want to create a new stretched vSAN compute cluster, see [Create a vSAN Compute Stretch Cluster Using the SDDC Manager API](#).

1. Create a JSON specification file in a text editor.

```
{
  "clusterStretchSpec": {
    "deployWithoutLicenseKeys": true,
    "hostSpecs": [
      {
        "id": "<ESX host 1 ID>",
        "hostname": "sfo02-w01-r01-esx01.sfo.rainpole.io",
        "hostNetworkSpec": {
          "networkProfileName": "sfo02-w01-r01-network-profile01",
          "vmNics": [
            {
              "id": "vmnic0",
              "vdsName": "sfo-w01-cl01-vds01",
              "uplink": "uplink1"
            },
            {
              "id": "vmnic1",
              "vdsName": "sfo-w01-cl01-vds01",
              "uplink": "uplink2"
            }
          ]
        }
      },
      {
        "id": "<ESX host 2 ID>",
        "hostname": "sfo02-w01-r01-esx02.sfo.rainpole.io",
        "hostNetworkSpec": {
          "networkProfileName": "sfo02-w01-r01-network-profile01",
          "vmNics": [
            {
              "id": "vmnic0",
              "vdsName": "sfo-w01-cl01-vds01",
              "uplink": "uplink1"
            }
          ]
        }
      }
    ]
  },
  "networkSpec": {
    "networkProfiles": [
      {
        "isDefault": true,
        "name": "sfo02-w01-r01-network-profile01",
        "nsxtHostSwitchConfigs": [
          {
            "ipAddressPoolName": "sfo02-w01-r01-ip-pool01-host",
            "uplinkProfileName": "sfo02-w01-r01-uplink-profile01",
            "vdsName": "sfo-w01-cl01-vds01",
            "vdsUplinkToNsxUplink": [
              {
                "nsxUplinkName": "uplink1",
                "vdsUplinkName": "uplink1"
              },
              {
                "nsxUplinkName": "uplink2",
                "vdsUplinkName": "uplink2"
              }
            ]
          }
        ]
      }
    ]
  },
  "nsxClusterSpec": {
```

```
{
  "id": "vmnic1",
  "vdsName": "sfo-w01-cl01-vds01",
  "uplink": "uplink2"
}
},
{
  "id": "<ESX host 3 ID>",
  "hostname": "sfo02-w01-r01-esx03.sfo.rainpole.io",
  "hostNetworkSpec": {
    "networkProfileName": "sfo02-w01-r01-network-profile01",
    "vmNics": [
      {
        "id": "vmnic0",
        "vdsName": "sfo-w01-cl01-vds01",
        "uplink": "uplink1"
      },
      {
        "id": "vmnic1",
        "vdsName": "sfo-w01-cl01-vds01",
        "uplink": "uplink2"
      }
    ]
  }
},
{
  "networkSpec": {
    "networkProfiles": [
      {
        "isDefault": true,
        "name": "sfo02-w01-r01-network-profile01",
        "nsxtHostSwitchConfigs": [
          {
            "ipAddressPoolName": "sfo02-w01-r01-ip-pool01-host",
            "uplinkProfileName": "sfo02-w01-r01-uplink-profile01",
            "vdsName": "sfo-w01-cl01-vds01",
            "vdsUplinkToNsxUplink": [
              {
                "nsxUplinkName": "uplink1",
                "vdsUplinkName": "uplink1"
              },
              {
                "nsxUplinkName": "uplink2",
                "vdsUplinkName": "uplink2"
              }
            ]
          }
        ]
      }
    ]
  },
  "nsxClusterSpec": {
```

```

"ipAddressPoolsSpec": {
  "name": "sfo02-w01-r01-ip-pool01-host",
  "subnets": [
    {
      "cidr": "10.14.14.0/24",
      "gateway": "10.14.14.1",
      "ipAddressPoolRanges": [
        {
          "start": "10.14.14.101",
          "end": "10.14.14.132"
        }
      ]
    }
  ]
},
"uplinkProfiles": [
  {
    "name": "sfo02-w01-r01-uplink-profile01",
    "transportVlan": 1414,
    "teamings": [
      {
        "name": "DEFAULT",
        "policy": "LOADBALANCE_SRCID",
        "standByUplinks": [],
        "activeUplinks": [
          "uplink1",
          "uplink2"
        ]
      }
    ]
  }
]
},
"isEdgeClusterConfiguredForMultiAZ": true
}

```

Note: Replace the example values in the JSON file with the correct values for your environment.

2. Log in to VCF Operations.
3. In the navigation pane, click **Developer Center > APIs & SDKs**.
4. Click the **API Explorer** link on the SDDC Manager API.
5. Click the VCF Instance which contains the cluster you want to stretch.
6. Retrieve the unique IDs for each ESX host that will be part of the stretched cluster and add the IDs to the JSON specification file.
 - a) Navigate to **Hosts > GET /v1/hosts**.
 - b) In the **Status** text box, enter `UNASSIGNED_USEABLE` and click **Execute**.
 - c) In the **Response** section, click **PageOfHost**, copy the `id` element of each host, and replace the respective value in the JSON specification file.

ESX Host	Value
ESX Host 1	<i>ESX host 1 ID</i>
ESX Host 2	<i>ESX host 2 ID</i>
ESX Host 3	<i>ESX host 3 ID</i>

7. Retrieve the unique ID for the cluster you want to stretch.
 - a) Navigate to **Clusters > GET /v1/clusters**.
 - b) Click **Execute**.
 - c) In the **Response** section, click **PageOfCluster**, copy the `id` element of the cluster.
You will need the cluster ID for the subsequent step.
8. Stretch the cluster with the JSON specification.
 - a) Navigate to **Clusters > PATCH /v1/clusters/{id}**.
 - b) Paste the unique `ID` of the cluster in the **Value** text-box.
 - c) In the **clusterUpdateSpec** text box, paste the JSON specification.
 - d) Click **Execute**.
 - e) On the confirmation dialog box, click **Continue**.
You can monitor the stretch cluster creation in **Fleet Management > Tasks**.

Stretch a vSAN Storage Cluster Using the SDDC Manager API

You can stretch a vSAN storage cluster using the `PATCH /v1/clusters/{id}` API. You must stretch a vSAN storage server cluster before you can stretch any of its vSAN compute client clusters.

- See [vSAN Stretched Storage Cluster Storage Model](#) for design requirements and recommendations.
- Commission a minimum of three ESX hosts with the vSAN type vSAN Storage (formerly known as vSAN Max). See [Commission ESX Hosts](#).
- An existing vSAN storage cluster with a minimum of three ESX hosts.
- Deploy and configure a vSAN ESA witness host. See [Deploy and Configure vSAN Witness Host](#).
- Unmount any vSAN compute cluster datastores that are mounted to the vSAN storage cluster.

1. Create a JSON specification file in a text editor.

The following example is for an environment with two vSphere Distributed Switches. This is typical in a vSAN storage cluster where vSAN storage traffic separation is configured with a dedicated vDS just for the vSAN storage traffic and all other traffic including the vSAN client traffic is on a separate vDS.

Note:

The ESX hosts that you are adding to availability zone 2 must use the same vmnic to vSphere Distributed Switch mapping as the existing hosts in availability zone 1.

```
{
  "clusterStretchSpec": {
    "hostSpecs": [
      {
        "id": "<ESX host 1 ID>",
        "hostname": "sfo02-w01-r01-esx01.sfo.rainpole.io",
        "hostNetworkSpec": {
          "networkProfileName": "sfo02-w01-r01-network-profile01",
          "vmNics": [
            {
              "id": "vmnic0",
              "vdsName": "sfo-w01-cl01-vds01",
              "uplink": "uplink1"
            },
            {
              "id": "vmnic1",
              "vdsName": "sfo-w01-cl01-vds01",
              "uplink": "uplink2"
            },
            {
              "id": "vmnic2",
              "vdsName": "sfo-w01-cl01-vds02",
              "uplink": "uplink1"
            },
            {
              "id": "vmnic3",
              "vdsName": "sfo-w01-cl01-vds02",
              "uplink": "uplink2"
            }
          ]
        }
      },
      {
        "id": "<ESX host 2 ID>",
        "hostname": "sfo02-w01-r01-esx02.sfo.rainpole.io",
        "hostNetworkSpec": {
          "networkProfileName": "sfo02-w01-r01-network-profile01",
          "vmNics": [
            {
              "id": "vmnic0",
              "vdsName": "sfo-w01-cl01-vds01",
              "uplink": "uplink1"
            },
            {

```

```
        "id": "vmnic1",
        "vdsName": "sfo-w01-cl01-vds01",
        "uplink": "uplink2"
      },
      {
        "id": "vmnic2",
        "vdsName": "sfo-w01-cl01-vds02",
        "uplink": "uplink1"
      },
      {
        "id": "vmnic3",
        "vdsName": "sfo-w01-cl01-vds02",
        "uplink": "uplink2"
      }
    ]
  },
  {
    "id": "<ESX host 3 ID>",
    "hostname": "sfo02-w01-r01-esx03.sfo.rainpole.io",
    "hostNetworkSpec": {
      "networkProfileName": "sfo02-w01-r01-network-profile01",
      "vmNics": [
        {
          "id": "vmnic0",
          "vdsName": "sfo-w01-cl01-vds01",
          "uplink": "uplink1"
        },
        {
          "id": "vmnic1",
          "vdsName": "sfo-w01-cl01-vds01",
          "uplink": "uplink2"
        },
        {
          "id": "vmnic2",
          "vdsName": "sfo-w01-cl01-vds02",
          "uplink": "uplink1"
        },
        {
          "id": "vmnic3",
          "vdsName": "sfo-w01-cl01-vds02",
          "uplink": "uplink2"
        }
      ]
    }
  },
  "networkSpec": {
    "networkProfiles": [
      {
        "isDefault": true,
        "name": "sfo-w01-r01-network-profile-01",
        "nsxtHostSwitchConfigs": [

```

```

    {
      "ipAddressPoolName": "sfo02-w01-r01-ip-pool01-host",
      "uplinkProfileName": "sfo02-w01-r01-uplink-profile01",
      "vdsName": "sfo-w01-cl01-vds01",
      "vdsUplinkToNsxUplink": [
        {
          "nsxUplinkName": "uplink1",
          "vdsUplinkName": "uplink1"
        },
        {
          "nsxUplinkName": "uplink2",
          "vdsUplinkName": "uplink2"
        }
      ]
    }
  ]
},
"nsxClusterSpec": {
  "ipAddressPoolsSpec": [
    {
      "name": "sfo02-w01-r01-ip-pool01-host",
      "subnets": [
        {
          "cidr": "10.14.14.0/24",
          "gateway": "10.14.14.1",
          "ipAddressPoolRanges": [
            {
              "start": "10.14.14.101",
              "end": "10.14.14.132"
            }
          ]
        }
      ]
    }
  ]
},
"uplinkProfiles": [
  {
    "name": "sfo02-w01-r01-uplink-profile-01",
    "transportVlan": 1414,
    "teamings": [
      {
        "name": "DEFAULT",
        "policy": "LOADBALANCE_SRCID",
        "standByUplinks": [],
        "activeUplinks": [
          "uplink1",
          "uplink2"
        ]
      }
    ]
  }
]
}
]

```

```

    }
  },
  "isEdgeClusterConfiguredForMultiAZ": false,
  "witnessSpec": {
    "fgdn": "sfo-w01-cl01-vsw01.sfo.rainpole.io",
    "vsanCidr": "10.21.10.0/24",
    "vsanIp": "10.21.10.219"
  },
  "witnessTrafficSharedWithVsanTraffic": false,
  "deployWithoutLicenseKeys": true
}
}

```

Note: Replace the example values in the JSON file with the correct values for your environment.

- Log in to VCF Operations.
- In the navigation pane, click **Developer Center > APIs & SDKs**.
- Click the **API Explorer** link on the SDDC Manager API.
- Click the VCF Instance which contains the cluster you want to stretch.
- Retrieve the unique IDs for each ESX host that will be part of the stretched cluster and add the IDs to the JSON specification file.
 - Navigate to **Hosts > GET /v1/hosts**.
 - In the **Status** text box, enter `UNASSIGNED_USEABLE` and click **Execute**.
 - In the **Response** section, click **PageOfHost**, copy the `id` element of each host, and replace the respective value in the JSON specification file.

ESX Host	Value
ESX Host 1	<i>ESX host 1 ID</i>
ESX Host 2	<i>ESX host 2 ID</i>
ESX Host 3	<i>ESX host 3 ID</i>

- Retrieve the unique ID for the cluster you want to stretch.
 - Navigate to **Clusters > GET /v1/clusters**.
 - Click **Execute**.
 - In the **Response** section, click **PageOfCluster**, copy the `id` element of the cluster. You will need the cluster ID for the subsequent step.
- Stretch the cluster with the JSON specification.
 - Navigate to **Clusters > PATCH /v1/clusters/{id}**.
 - Paste the unique ID of the cluster in the **Value** text-box.
 - In the **clusterUpdateSpec** text box, paste the JSON specification.
 - Click **Execute**.
 - On the confirmation dialog box, click **Continue**.
You can monitor the stretch cluster creation in **Fleet Management > Tasks**.

Expand a Stretched Cluster in VMware Cloud Foundation

You can expand a stretched cluster by adding ESX hosts. It is recommended that you add the same number of hosts to both availability zones for symmetry and cluster balance.

1. Commission the additional hosts to the global inventory.

See [Commission ESX Hosts](#).

Note: Make sure to select the correct network pool for the availability zone to which you are adding each host.

2. Get the UUIDs of the hosts you commissioned.
 - a) In the navigation pane, click **Developer Center > API Explorer**.
 - b) Under APIs for managing hosts, click **GET /v1/hosts**.
 - c) Click **Execute**.
 - d) Click **Download** to download the JSON file.
 - e) Open the JSON file and copy the the UUIDs of the hosts.
3. Get the ID of the cluster you are expanding.
 - a) In the API Explorer, navigate to APIs for managing clusters and click GET /v1/clusters.
 - b) Click **Execute**.
 - c) Click **Download** to download the JSON file.
 - d) Open the JSON file and copy the the cluster ID for the cluster you are expanding.
4. Get the primary and secondary availability zone names from vCenter.
 - a) In a web browser, log in to the vCenter at https://vcenter_fqdn/ui.
 - b) Select **Menu > Hosts and Clusters**.
 - c) In the inventory panel, expand **vCenter > Datacenter**.
 - d) Select the cluster and then click the **Configure** tab.
 - e) Under **vSAN**, select **Fault Domains**.
 - f) Note the primary and secondary availability zone names.
5. Prepare the JSON request body.
 - a) Click **Patch /v1/clusters/id**.
 - b) Under **ClusterUpdateSpec** field, click **Cluster Update Data ClusterUpdateSpec{ ... }**.
 - c) Click **Download** to download the JSON file.
 - d) Edit the downloaded JSON file so that it contains only the expand section similar to the example below. In the **azName** field, type the primary and secondary names you had retrieved in step 4.

Note: The ESXi hosts that you are adding must use the same vmnic to vSphere Distributed Switch mapping as the existing hosts in the stretched cluster. For example: If existing hosts map vmnic0 and vmnic1 to vSphere Distributed Switch 1 and vmnic2 and vmnic3 to vSphere Distributed Switch 2, then the hosts you are adding must map the same vmnics to the same vSphere Distributed Switches.

```
{
  "clusterExpansionSpec": {
    "hostSpecs": [ {
      "id": "ESX host 1 ID",
      "azName": "primary/secondary",
      "hostNetworkSpec": {
        "vmNics": [ {
          "id": "vmnic0",
          "vdsName": "<vSphere Distributed Switch 1>"
        },
        {
          "id": "vmnic1",
          "vdsName": "<vSphere Distributed Switch 2>"
        }
      ]
    },
    {
      "id": "ESX host 2 ID",
      "azName": "primary/secondary",
      "hostNetworkSpec": {
        "vmNics": [ {
          "id": "vmnic0",
          "vdsName": "<vSphere Distributed Switch 1>"
        },
        {
          "id": "vmnic1",
          "vdsName": "<vSphere Distributed Switch 2>"
        }
      ]
    }
  ]
}
```

```
    "deployWithoutLicenseKeys": "true"
  }, {
    "id": "ESX host 2 ID",
    "azName": "primary/secondary",
    "hostNetworkSpec": {
      "vmNics": [ {
        "id": "vmnic0",
        "vdsName": "<vSphere Distributed Switch 1>"
      },
      {
        "id": "vmnic1",
        "vdsName": "<vSphere Distributed Switch 2>"
      }
    ]
  },
  "deployWithoutLicenseKeys": "true"
}
], {
  "id": "ESX host 3 ID",
  "azName": "primary/secondary",
  "hostNetworkSpec": {
    "vmNics": [ {
      "id": "vmnic0",
      "vdsName": "<vSphere Distributed Switch 1>"
    },
    {
      "id": "vmnic1",
      "vdsName": "<vSphere Distributed Switch 2>"
    }
  ]
},
"deployWithoutLicenseKeys": "true"
}
], {
  "id": "ESX host 4 ID",
  "azName": "primary/secondary",
  "hostNetworkSpec": {
    "vmNics": [ {
      "id": "vmnic0",
      "vdsName": "<vSphere Distributed Switch 1>"
    },
    {
      "id": "vmnic1",
      "vdsName": "<vSphere Distributed Switch 2>"
    }
  ]
},
"deployWithoutLicenseKeys": "true"
}
} ]
}
```

6. Run the expand cluster API.
 - a) For the **ClusterUpdateSpec** field, update the cluster ID (you retrieved this in step 3) and JSON file with the payload you prepared in step 5.
 - b) Click **Execute**.
 - c) Monitor the task until it is completed.
7. If required, SSH in to each newly added host and add a static route to the vSAN network of the witness host. Also add static routes in the witness if it could not reach the vSAN network of the newly added hosts.
8. Update the value of **Host failure cluster tolerates** to the number of hosts in AZ1 after cluster expansion.
 - a) Log in to the management vCenter instance.
 - b) Select **Cluster** and click the **Configure** tab.
 - c) Under **Services**, click **vSphere Availability** and then click **Edit**.
 - d) On the **Admission Control** page of the **Edit Cluster Settings** dialog box, set host failures cluster tolerates to the number of hosts in availability zone 1 and click **OK**.

Unstretch a Cluster in VMware Cloud Foundation

This procedure describes how to unstretch a vSAN cluster in VMware Cloud Foundation.

- Consolidate or delete all virtual machine snapshots on the vSAN cluster before unstretching the cluster.
 - If you are unstretching a vSAN storage cluster (formerly known as vSAN Max), unmount any stretched vSAN compute clusters to which the vSAN storage cluster is providing a remote datastore.
1. Get the ID of the cluster you are unstretching.
 - a) In the VCF Operations console, click **Developer Center > APIs & SDKs**.
 - b) Click the **API Explorer** link for the SDDC Manager API.
 - c) In the API Explorer, navigate to APIs for managing clusters and click `GET /v1/clusters`.
 - d) Click **Execute**.
 - e) Click **Download** to download the JSON file.
 - f) Open the JSON file and copy the the cluster ID for `SDDC-Cluster1`.
 2. Prepare the JSON request body.
 - a) Click **Patch /v1/clusters/id**.
 - b) Under **ClusterUpdateSpec** field, click **Cluster Update Data ClusterUpdateSpec{ ... }**.
 - c) Click **Download** to download the JSON file.
 - d) Edit the downloaded JSON file so that it contains only the unstretch information similar to the example below.


```
{ "clusterUnstretchSpec": {} }
```
 3. Run the unstretch cluster API.
 - a) For the **ClusterUpdateSpec** field, update the cluster UID (you retrieved this in step 1) and unstretch JSON file with the payload you prepared in step 2.
 - b) Click **Execute**.
The unstretch cluster task is displayed in the SDDC Manager task panel.
 - c) Monitor the unstretch cluster task till it is completed.
All hosts from AZ2 are removed from the unstretched cluster and the cluster is converted to standard vSAN cluster.

Replace a Failed Host in a Stretched Cluster

If an ESX host or ESX host component in a stretched cluster fails, it is recommended that you replace the host with a new host.

- Image the replacement host with the same ESX version as the other hosts in the cluster.

- Check the health of the cluster.
1. Get the IDs of the hosts to be removed.
 - a) In the VCF Operations console, click **Developer Center > APIs & SDKs**.
 - b) Click the **API Explorer** link for the SDDC Manager API.
 - c) Under APIs for managing hosts, click **GET /v1/hosts**.
 - d) Click **Execute**.
 - e) Click **Download** to download the JSON file.
 - f) Open the JSON file and copy the the ID of the host to be removed.
 2. Get the ID of the cluster from where the host is to be removed.
 - a) In the API Explorer, navigate to APIs for managing clusters and click `GET /v1/clusters`.
 - b) Click **Execute**.
 - c) Click **Download** to download the JSON file.
 - d) Open the JSON file and copy the the cluster ID.
 3. Prepare the JSON request body.
 - a) Click **Patch /v1/clusters/id**.
 - b) Under **ClusterUpdateSpec**, click **Cluster Update Data ClusterUpdateSpec{ ... }**.
 - c) Click **Download** to download the JSON file.
 - d) Edit the JSON file so that it contains only the compact section similar to the example below.

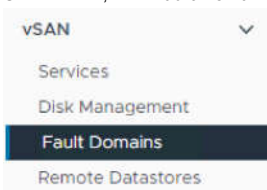

```
{
  "clusterCompactionSpec": {
    "hosts": [ {
      "id": "ESX host 1 ID"
    }, {
      "id": "ESX host 2 ID"
    }, {
      "id": "ESX host 3 ID"
    } ]
  }
}
```
 4. Run the compact cluster API.
 - a) In the `id` field, replace the values with the host IDs you retrieved in step 1.
 - b) Click **Execute**.
 - c) Monitor the task till it is completed.
 5. Decommission the host to be removed.
See [Decommission ESX Hosts](#).
 6. Commission the replacement host to the same network pool as the removed host.
See [Commission ESX Hosts](#).
 7. Expand the cluster to add the commissioned host to the cluster. See [Expand a Stretched Cluster in VMware Cloud Foundation](#).
 8. If required, SSH in to each newly added host and add a static route to the vSAN network of the witness host.

Change the vSAN Witness Host in a Stretched Cluster

You can replace or change the vSAN witness host for a stretched cluster using the vSphere Client without affecting lifecycle operations.

Verify that the vSAN witness host is not in use by another cluster, has a VMkernel configured for vSAN traffic, and has no vSAN partitions on its disks.

1. In a web browser, log in to vCenter at https://vcenter_fqdn/ui.
2. Select **Menu > Hosts and Clusters**.
3. In the inventory panel, expand **vCenter > Datacenter**.
4. Select the stretched cluster and click **Configure**.
5. Under vSAN, click **Fault Domains**.



6. Click the **Change** button.
7. Select a new host to use as the vSAN witness host and click **Next**.
8. Claim disks on the new witness host and click **Next**.
9. Review the configuration and click **Finish**.

Managing Network Connectivity in vCenter

Starting with VCF 9.0, a simplified workflow is introduced to set up network connectivity and perform network configuration.

The new NSX-based workflow for network connectivity is available both in vCenter and NSX for you to have a uniform experience of performing network configurations from either of the components. The new workflow leverages many default settings that are recommended by Broadcom for simplifying the deployment process, such as deployment parameters, transport zones, and uplink profile, that reduces the number of user inputs required to perform network configuration. The workflow also provides a topology visualization to correlate your inputs in the user interface. Contextual visualization helps you get a view of the set up before completing the tasks in the workflow.

Based on the kind of network connectivity you have selected, you can create an edge cluster with a centralized gateway or a distributed gateway with VLAN. For more information, see [Workload Networking Detailed Designs](#) and [NSX Edge Cluster Detailed Design](#).

Configure Centralized Network Connectivity with Edge Clusters

Create this connection if you have an environment that requires static or dynamic routing with physical fabric. A centralized connection is the one where the traffic is routed through a NSX gateway on an edge node. You can use this task to configure a centralized connectivity with edge cluster deployment from vCenter. For more information on network connectivity using NSX, see [Setting up Centralized Connectivity with Edge Clusters](#).

If you want to deploy edges on a vSphere cluster, ensure that the following requirements are met. VCF Installer and VCF Operations fulfill a number of prerequisites by default.

- The vSphere cluster must have ESX hosts with vSphere Distributed Switch uplinks configured.
- If you are using DHCP to allocate MGMT or TEP IP addresses, then create and configure a DHCP scope accordingly.
- The Compute Manager must be registered with NSX Manager.

- Plan for Edge TEP VLAN. The same VLAN as the Host TEP can be reused. For more information on creating TEPs, see [Create an IP Pool for Host Tunnel Endpoint IP Addresses](#).
- Edges can be deployed on a host-group within a vSphere cluster or on a vSphere cluster. All hosts in the host-group or the vSphere cluster must have identical access to management network, gateway uplinks, host and Edge TEP networks.
- The vSphere cluster must have a Transport Node Profile (TNP) attached to it. For more information on applying a TNP, see [Prepare Cluster Hosts as Transport Nodes by Using TNP](#).
- If you want to use the same VLAN for edge and host overlay, then, ensure that the **Activate NSX on DVP** option is turned on for the host cluster in which the edge is to be deployed. To turn it on, go to **System > Fabric > Hosts** page for cluster in NSX Manager.
- Edge node will reserve 100% CPU shares and memory. vSphere cluster and resource pool must have enough resources to deploy at least 2 edge nodes.
- NSX Edge VM Resource Requirements:
 - **Small**: 4 GB memory, 2 vCPU, 200 GB disk space. The NSX Edge Small VM appliance size is suitable for lab and proof-of-concept deployments.
 - **Medium**: 8 GB memory, 4 vCPU, 200 GB disk space. The NSX Edge Medium appliance size is suitable for production environments with load balancing.
 - **Large**: 32 GB memory, 8 vCPU, 200 GB disk space. The NSX Edge Large appliance size is suitable for production environments with load balancing.
 - **XLarge**: 64 GB memory, 16 vCPU, 200 GB disk space. The NSX Edge Extra Large appliance size is suitable for production environments with load balancing.
- Management network for NSX Edge must have connectivity to NSX Manager.
- DNS entries for NSX Edge components should be populated in your managed DNS server.
- If dynamic routing is desired, please set up 2 BGP Peers on ToR switch with an interface IP, local ASN, and BGP password. For more information about configuring BGP, see [Configure BGP](#).
- If you are using Border Gateway Protocol (BGP), reserve a local ASN to use for the NSX Edge cluster's Tier-0 Interfaces.
- If static routing is desired, it is recommended to configure HA VIP for active/standby Tier-0 Gateway and use VIP as the Next Hop IP on Top-of-Rack (ToR) switch.
- When setting up the centralized connectivity by using the simplified workflow in vCenter, you must add a minimum of two NSX Edge nodes to a NSX Edge cluster. You can add up to 10 edges to an edge cluster. Once a NSX Edge cluster is successfully created, you will see the NSX Edge cluster list on subsequent visits to the **Network Connectivity** page.
- For more information on planning resources, see [Planning and Preparation Workbook](#).

If you have upgraded to VCF 9.0, you can use this workflow to manage edge clusters created in the previous version. Additionally, you can also use this workflow to set up a Tier-0 gateway for an edge cluster, add or delete an edge, and reset passwords. Based on these configurations, you can start creating centralized gateway connectivity with edge cluster deployment.

1. In the *vSphere Client*, navigate to the vCenter for the Workload Domain and select **Networks > Network Connectivity**.
2. Click **Configure Network Connectivity**.
3. In the **Gateway Type** page, select **Centralized Connectivity**.
4. Click **Next**.
5. Review and complete the **Networking Prerequisites**.
6. Click **Continue**.
7. Configure an Edge Cluster.
 - a) Enter a name for the edge cluster.
 - b) Enter the value for Tunnel Endpoint MTU. The maximum packet size allowed over a tunnel is already selected from the Global Fabric Settings.
Note:
The Edge TEP MTU is the same across the host MTU within the same overlay Transport Zone, so you must set the Global Fabric MTU accordingly.
8. Define the settings for edge nodes.
 - a) Select an appropriate size for edge cluster in **Edge Form Factor**.
 - b) A minimum of 2 Edge Nodes are required to deploy an Edge Cluster. Once you have added a node, you can add another edge node by cloning the previously added edge node. Cloning is recommended only when the cloned Edge is part of the same vSphere cluster with identical networks (edge management, pNIC mapping, VLAN for Edge TEP).
 - c) If the **Manage Edge Password with VCF** is turned on, then passwords are generated by with VCF to manage Edge nodes. If the **Manage Edge Password with VCF** is turned off, then you can manually set the password.
9. Manage passwords.
 - a) If the **Manage Edge Password with VCF** is turned on, then password is generated by with VCF to manage Edge nodes. If the **Manage Edge Password with VCF** is turned off, then you can manually set the password.
 - b) Enter the CLI and Root Administrator passwords to manage the Edge Nodes.
Your passwords must comply with the password strength restrictions.
 - At least 12 characters
 - At least one lower-case letter
 - At least one upper-case letter
 - At least one digit
 - At least one special character
 - At least five different characters

Option	Description
CLI Username	Enter the name for the CLI user.
CLI password	Enter the password for the CLI user.
CLI Confirm Password	Enter the password again for confirmation.
System Root password	Enter the password for the root admin user.
System Root Confirm Password	Enter the password again for confirmation.

10. To add a NSX Edge Node to a NSX Edge Cluster, click **Add** to add an **Edge Node** in the **Edge Cluster** page.

Option	Description
Edge Node Name (FQDN)	Enter the FQDN for the NSX Edge node. Each node must have a unique FQDN.
vSphere Cluster	Select a vSphere cluster to host the NSX Edge node. You can select a standard vSphere cluster or a stretched vSphere cluster, but all the NSX Edge nodes in an NSX Edge cluster must be hosted on vSphere clusters of the same type. If the vSphere cluster you select already hosts management virtual machines that are connected to the host Management port group, the VM Management Portgroup VLAN settings are not available.
Resource Pool (Optional)	Select the resource pool for the Edge node. Note: Resource Pool must be created before running the workflow.
Host Group Affinity	Enter Yes if you want the NSX Edge to be pinned to a host group affinity. Edge host affinity enables the Edge to be pinned to a given host or group of hosts. Best Effort Restart (BER) policy is applied to the Edge VMs that ensures a graceful shutdown of the Edge VM when host enters maintenance mode.
Host Group	If you have selected the Host Group Affinity as Yes , then select a host group to which the Edge node needs to be pinned. Note: Host group must be created before running the workflow.
Data Store	Select a data store where the Edge node virtual machine will be placed.
Management IP	
IP Allocation	Select DHCP or Static.
Port Group	Select the port group for the node. • For management workload, VM management port group is created. If a different port group is required, then create the port group before performing this operation. • For a VI workload, VM management port group is not created. Create a port group before performing this operation.
Management IP CIDR	Enter a CIDR for Management IP if you have selected Static option for IP allocation. Each node must have a unique management IP.
Default Gateway	Enter the default gateway address if you selected the Static option for IP allocation.
Uplinks	If Use the host overlay network configuration from the selected vSphere cluster checkbox is selected, the existing overlay network configuration (such as VLAN ID and TEP IP assignment) is automatically fetched from the hosts in the selected vSphere cluster. This option will be enabled if the selected cluster/host is prepared and NSX enabled on DVP setting is Yes . If this checkbox is not selected, manually enter the information for uplink configuration.

Option	Description
Edge Node Uplink Mapping	ESX is added to atleast one VDS by which ESX pNICs get mapped to uplinks on VDS. Map the Edge NIC to these ESX pNICs. The pNICs are also auto-selected when there are two pNICs on a VDS. In this case, if you choose the first pNIC as an active uplink for the first interface (fp-eth0) the other pNIC is automatically selected as a standby uplink. For the second interface (fp-eth1), exactly the reverse pNICs are selected as active and standby uplinks.If there are multiple pNICs or pNICs connected to multiple VDS, pNICs are not auto-selected and you will have to manually map them.
TEP VLAN	Select the VLAN ID for Tunnel Endpoints (TEP). This field will be auto populated if the Use the host overlay network configuration from the selected vSphere cluster checkbox is selected.
IP Allocation (TEP)	Select the IP allocation method for TEP. Define a vLAN ID for the Host TEP network. You can select from the following options. Based on the IP allocation method, provide the other required details. - DHCP - DHCP Server provides IP addresses to the hosts. The TEP IP is allocated by the external DHCP server. - IP Pool - Select an IP Pool for allocations. - Static IP List - Provide values IPv4 Static List, Static IPv4 Gateway, and IPv4 Subnet Mask. This field will be auto populated if the Use the host overlay network configuration from the selected vSphere cluster checkbox is selected.
IP Pool	Select the IP pool from the drop-down menu.

11. Under **Uplinks**, select **Use the host overlay network configuration from the selected vSphere Cluster**. When you enable this, it automatically gets the existing overlay network configuration from the hosts in the selected vSphere cluster.

12. Click **Apply**

13. In the **Edge Cluster** page, use **Clone** to clone the Edge Node.

14. Configure the settings that cannot be cloned.

Option	Description
Edge Node Name (FQDN)	Enter the FQDN for the NSX Edge node. Each node must have a unique FQDN.

15. Click **Apply**.

16. In the **Workload Domain Connectivity** page, configure the parameters to enable external connectivity.

17. To skip gateway and routing configurations for now and configure them later, click **Skip**. In this case, the system will hide the Gateway and Routing fields and the edges will be deployed without the gateway and routing configurations. You can configure these settings later from vCenter. Navigate to the vCenter instance for the management or workload domain and select **Networks > Network Connectivity**. On the **Edge Cluster** page, click **Set** under **Gateway**.

Option	Description
Gateway Name	Enter a name for the gateway.

Option	Description
High Availability Mode	Select the HA mode for the gateway.
Routing Configuration	
Gateway Routing Type	Select from BGP or Static options.
Local Autonomous System	Enter the autonomous system number if the gateway routing type is set to BGP.

18. Set **Gateway Uplinks**. For each edge node, configure your gateway uplink to establish an uplink connectivity between gateway and Top of the Rack router. You can only configure up to two Edge Nodes with Gateway uplinks if HA mode is set to Active Standby.

Option	Description
Gateway Interface VLAN	Enter the gateway interface vLAN. This field is displayed when Gateway Routing Type is set to Static .
Gateway Interface CIDR	Enter the gateway IP address. This field is displayed when Gateway Routing Type is set to Static .
BGP Peer IP	Enter first uplink BGP peer IP to establish a connection between a Tier-0 gateway and a router in your physical infrastructure. This field is displayed when Gateway Routing Type is set to BGP .
MTU	Enter the MTU for the NSX Edge cluster. The MTU value can be between 1500-9000. This field is displayed when Gateway Routing Type is set to BGP .
BGP Peer Password	Enter BGP password. This field is displayed when Gateway Routing Type is set to BGP .
Confirm Password	Confirm the BGP password. This field is displayed when Gateway Routing Type is set to BGP .

19. To enable networking inside and outside VPC, IP blocks must be pre-provisioned where transit gateway is defined. Under VPC configuration, set the parameters.

To edit this configuration in the *vSphere Client*, navigate to the vCenter instance for the management or workload domain and select **Networks > Network Connectivity**.

Option	Description
External IP Blocks	Enter external IP blocks that can be used to create public subnets in the VPC. The entered IP blocks allow outside connectivity to VPC workloads either on public subnets or through external IPs.
Private - Transit Gateway IP Blocks	Enter private IP blocks to create private subnets in the VPC. The entered IP blocks are available for inter-VPC communication and are not advertised by the Transit Gateway to the outside datacenter.

20. Review your configurations under **Edge Cluster Details** and **Workload Domain Connectivity**.

21. Click **Deploy**.

Add an NSX Edge Node to an NSX Edge Cluster

Learn how to add a NSX Edge Node to a Edge Cluster through vCenter. It provides a simplified workflow, centralized management, reduced complexity, and improved operational efficiency for virtual private cloud environments.

When setting up the centralized connectivity by using the simplified workflow in vCenter, you must add a minimum of two Edge nodes to an Edge cluster. You can add up to 10 edges to an edge cluster. Once an Edge cluster is successfully created, you will see an Edge cluster list on subsequent visits to the **Network Connectivity** page.

1. In the *vSphere Client*, navigate to the vCenter for the management or workload domain and select **Networks > Network Connectivity**.
2. On the **Edge Cluster** page, click **Add** to add an **Edge Node**.

Option	Description
Edge Node Name (FQDN)	Enter the FQDN for the NSX Edge node. Each node must have a unique FQDN.
vSphere Cluster	Select a vSphere cluster to host the NSX Edge node. You can select a standard vSphere cluster or a stretched vSphere cluster, but all the NSX Edge nodes in an NSX Edge cluster must be hosted on vSphere clusters of the same type. If the vSphere cluster you select already hosts management virtual machines that are connected to the host Management port group, the VM Management Portgroup VLAN settings are not available.
Resource Pool (Optional)	Select the resource pool for the Edge node. Note: Resource Pool must be created before running the workflow.
Host Group Affinity	Enter Yes if you want the Edge to be pinned to a host group affinity. Edge host affinity enables the Edge to be pinned to a given host or group of hosts. Best Effort Restart (BER) policy is applied to the Edge VMs that ensures a graceful shutdown of the Edge VM when host enters maintenance mode.
Host Group	If you have selected the Host Group Affinity as Yes , then select a host group to which the Edge node needs to be pinned. Note: Host group must be created before running the workflow.
Data Store	Select a data store where the Edge node virtual machine will be placed.
Management IP	
IP Allocation	Select DHCP or Static.
Port Group	Select the port group for the node. • For management workload, VM management port group is created. If a different port group is required, then create the port group before performing this operation. • For a VI workload, VM management port group is not created. Create a port group before performing this operation.

Option	Description
Management IP CIDR	Enter a CIDR for Management IP if you have selected Static option for IP allocation. Each node must have a unique management IP.
Default Gateway	Enter the default gateway address if you selected the Static option for IP allocation.
Uplinks	If Use the host overlay network configuration from the selected vSphere cluster checkbox is selected, the existing overlay network configuration (such as VLAN ID and TEP IP assignment) is automatically fetched from the hosts in the selected vSphere cluster. This option will be enabled if the selected cluster/host is prepared and NSX enabled on DVPG setting is Yes . If this checkbox is not selected, manually enter the information for uplink configuration.
Edge Node Uplink Mapping	ESX is added to atleast one VDS by which ESX pNICs get mapped to uplinks on VDS. Map the Edge NIC to these ESX pNICs. The pNICs are also auto-selected when there are two pNICs on a VDS. In this case, if you choose the first pNIC as an active uplink for the first interface (fp-eth0) the other pNIC is automatically selected as a standby uplink. For the second interface (fp-eth1), exactly the reverse pNICs are selected as active and standby uplinks. If there are multiple pNICs or pNICs connected to multiple VDS, pNICs are not auto-selected and you will have to manually map them.
TEP VLAN	Select the VLAN ID for Tunnel Endpoints (TEP). This field will be auto populated if the Use the host overlay network configuration from the selected vSphere cluster checkbox is selected.
IP Allocation (TEP)	Select the IP allocation method for TEP. Define a VLAN ID for the Host TEP network. You can select from the following options. Based on the IP allocation method, provide the other required details. • DHCP - DHCP Server provides IP addresses to the hosts. The TEP IP is allocated by the external DHCP server. • IP Pool - Select an IP Pool for allocations. • Static IP List - Provide values IPv4 Static List, Static IPv4 Gateway, and IPv4 Subnet Mask. This field will be auto populated if the Use the host overlay network configuration from the selected vSphere cluster checkbox is selected.
IP Pool	Select the IP pool from the drop-down menu.

3. Under **Uplinks**, select **Use the host overlay network configuration from the selected vSphere Cluster**. When you enable this, it automatically gets the existing overlay network configuration from the hosts in the selected vSphere cluster.
4. Click **Apply**.
5. In the **Edge Cluster** page, use **Clone** to clone the Edge Node.
6. Configure the settings that cannot be cloned. Enter the FQDN for the NSX Edge node. Each node must have a unique FQDN.
7. Click **Apply**.
8. In the **Workload Domain Connectivity** page, configure the parameters to enable external connectivity.
 - a) To skip gateway and routing configurations for now and configure them later, click **Skip**. In this case, the system will hide the Gateway and Routing fields and the edges will be deployed without the gateway and routing configurations. You can configure these settings later from vCenter. Navigate to the vCenter instance for the management or workload domain and select **Networks > Network Connectivity**. On the **Edge Cluster** page, click **Set** under **Gateway**.

Option	Description
Gateway Name	Enter the gateway name.
High Availability Mode	Select the HA mode for the gateway.
Routing Configuration	
Gateway Routing Type	Select from BGP or Static options.
Local Autonomous System	Enter the autonomous system number if the gateway routing type is set to BGP.

- b) Set **Gateway Uplinks**. For each edge node, configure your gateway uplink to establish an uplink connectivity between gateway and Top of the Rack router. You can only configure up to two Edge Nodes with Gateway uplinks if HA mode is set to Active Standby.

Option	Description
Gateway Interface VLAN	Enter the gateway interface vLAN. This field is displayed when Gateway Routing Type is set to Static .
Gateway Interface CIDR	Enter the gateway IP address. This field is displayed when Gateway Routing Type is set to Static .
BGP Peer IP	Enter first uplink BGP peer IP to establish a connection between a Tier-0 gateway and a router in your physical infrastructure. This field is displayed when Gateway Routing Type is set to BGP .
MTU	Enter the MTU for the NSX Edge cluster. The MTU value can be between 1500-9000. This field is displayed when Gateway Routing Type is set to BGP .
BGP Peer Password	Enter BGP password. This field is displayed when Gateway Routing Type is set to BGP .
Confirm Password	Confirm the BGP password.

Option	Description
	This field is displayed when Gateway Routing Type is set to BGP .

- c) To enable networking inside and outside VPC, IP blocks must be pre-provisioned where transit gateway is defined. Under VPC configuration, set the parameters. To edit this configuration in the *vSphere Client*, navigate to the vCenter instance for the management or workload domain and select **Networks > Network Connectivity**.

Option	Description
External IP Blocks	Enter external IP blocks that can be used to create public subnets in the VPC. The entered IP blocks allow outside connectivity to VPC workloads either on public subnets or through external IPs.
Private - Transit Gateway IP Blocks	Enter private IP blocks to create private subnets in the VPC. The entered IP blocks are available for inter-VPC communication and are not advertised by the Transit Gateway to the outside datacenter.

9. Review your configurations under **Edge Cluster Details** and **Workload Domain Connectivity**.

10. Click **Deploy**.

Configure Distributed Network Connectivity

Distributed connection allows a lightweight distributed connection to the data-center fabric without the need to deploy Edge nodes or configure dynamic routing. In addition to on-demand networking and distributed routing, other services available when using distributed connections are External IPs (1:1 NAT) and distributed DHCP.

- Dedicated VLAN(s) and subnets must be available for the Distributed Transit Gateway connection to the ToR. The same VLAN must be configured on all the ToR switches where the ESX host are connected which offers VPCs. All the hosts running VPC workloads must have access to these VLAN(s).
- Reserve external IP blocks for VPC connectivity. External IP blocks must be the same as or a subset of dedicated VLAN subnets. For more information, see [Create a Virtual Private Cloud in vCenter](#).

A distributed connection is the one where a Transit Gateway connects to a VLAN provisioned in the fabric. This happens directly on the host which implies the VLAN needs to be available on the *ESX* hosts (on physical uplinks used by the **NSX** vmkernel). The workflow is available in both **NSX** and vCenter. This section describes how to use the workflow from vCenter to configure a distributed connectivity. To set up a distributed gateway connectivity for the first time from vCenter, perform the following steps:

- From your browser, log in to the *vSphere Client* with administrator credentials.
- In the *vSphere Client*, navigate to the vCenter for the Workload Domain and select **Networks > Network Connectivity**.
- Click **Configure Network Connectivity**.
- In the **Transit Gateway** page, select **Distributed Connectivity**.

Select gateway connectivity for the created VPCs that align with the infrastructure requirements and networking preferences.

- Complete the **Networking prerequisites**. Click **Continue**.
- Enter the information on **External Network Connectivity Details** page.

Field Name	Description
VLAN ID	Enter the VLAN ID to which the traffic is to be routed. This VLAN must be dedicated for external connectivity of workloads deployment in VPCs and must be stretched across all clusters where VPCs are deployed.
Gateway CIDR IPv4 Address	Enter the gateway address through which the traffic should be routed to the external network.
VPC Configuration	
VPC External IPv4 Blocks	Enter IP blocks for VPC.
Private - TGW IPv4 Blocks	Enter IP blocks for Private - TGW IPV4.

- On the **Validations** page, review the details and click **Done**.

Add a VLAN Connection

Learn how to add a vLAN connection to your distributed network.

- In the *vSphere Client*, navigate to the vCenter for the Workload Domain and select **Networks > Network Connectivity**.
- Click **Add VLAN Connection** and add the information.

Option	Description
VPC External IPv4 Blocks	Enter IP blocks for VPC.
Private - TGW IPv4 Blocks	Enter IP blocks for Private - TGW IPV4.

- Click **Save**.

Managing Virtual Private Clouds in vCenter

Learn how to manage Virtual Private Clouds (VPC) in *vCenter*.

A virtual private cloud (VPC) is a configurable pool of shared network resources allocated within a cloud environment, providing a certain level of isolation between the different organizations using the resources. A subnet is a range of IP addresses in VPC and a workload is connected to a subnet.

In vSphere 8.0 Update 3, a hierarchical folder based approach was introduced to represent NSX projects and VPC in the *vSphere Client* that facilitates separate management by its various users. With this framework, NSX DVPGs were systematically organized in their respective folders. This streamlined the structural hierarchy between NSX and the *vCenter* inventory.

Prior to VCF 9.0, VPCs, subnets, and various VPC services were configured only from NSX. To connect the workloads, administrators had to navigate to *vCenter*. VCF 9.0 offers a simplified networking solution.

In VCF 9.0, VPCs and Subnets which are primarily NSX objects are integrated in *vCenter*. The administrator manages the VPC and subnet in *vCenter*. This implementation allows the VC admin to create, read, update and delete the NSX VPC from the vSphere Client. You can also navigate from and to NSX objects within the *vCenter*. For more information on VPCs in NSX, see [Virtual Private Cloud in NSX](#).

Create a Virtual Private Cloud in vCenter

Learn how to create a VPC in *vCenter*.

A VPC connects to a Transit Gateway (recommended) and can have the following types of subnets:

- Private - VPC:** A subnet scoped to the VPC and routed with other subnets inside the same VPC, but not to the outside network. NAT is needed if outside communication is required.
- Private - Transit Gateway:** Subnets allowing inter-VPC connectivity below the Transit Gateway without NAT. It requires IP translation if workloads need to be reachable to an outside network.
- Public:** A subnet that is accessible from outside the environment, from other VPCs but also from customers and workload above the Transit Gateway.

In case you have to grant access to specific workloads on private networks from the outside environment, you can use External IPs.

External IPs: IPs that allow outside connectivity for a VM on a private network by performing IP translation. It is implemented with 1:1 NAT on the workload picking an IP from External IP block and assigning it to the VM.

In addition to External IPs, it is also possible to configure Static Routes and Groups in a VPC.

Regardless of External Connection type (Centralized or Distributed), it is possible to configure Static Routes, Groups and DHCP Server in a VPC.

- From your browser, log in to the *vSphere Client*.
- In the *vSphere Client*, navigate to Virtual Private Clouds in the *vCenter* inventory. From the **Actions** drop-down menu, select **New VPC**.
- On the Basic Info page, enter the following information.

Option	Description
Name	Enter a name for the VPC.
Description	Enter a description for the VPC.
Private- VPC IP CIDRs	Enter a block of IP CIDRs that can be assigned to private subnets of this VPC. These CIDRs are local to VPCs and can overlap between the VPCs.
Connectivity & Services	You can modify these settings from the NSX UI. External IP Blocks: External IPs for the VMs attached to subnets. You will need to select the VM, the NIC, and whether to auto assign the IPs or provide IPs from an IP block. Private -Transit IP Blocks: Enter private IP blocks to create private subnets in the VPC. The entered IP blocks are available for inter-VPC communication. Networks created from the Private - Transit Gateway IP blocks are not advertised. Transit Gateway Connectivity: You can select the default connectivity profile and continue the VPC configuration based on the simplified VPC model. For more information about the VPC Connectivity Profile, see Add a VPC Connectivity Profile. N-S Services: Turn on the toggle to enable service gateway for the connected subnets and to support centralized services such as N-S firewall, NAT, or gateway QoS profile in the VPC. Default Outbound NAT: Turn on the toggle if the traffic from workloads on private subnets can be routed outside the NSX VPC. Note:

Option	Description
	This toggle is available only if N-S Services the toggle is turned on. DHCP: One of the following options is present: None: DHCP will not be enabled in the subnet and all IP addresses will be set statically. DHCP Server: The DHCP server specified in the service profile will automatically assign IP addresses and network settings to VMs connected to the subnet. You will also need to set Generic DHCP Options, and Static Bindings if you select this option. If the Auto allocate Subnet CIDR from IP Block toggle is off, then you need to provide IP CIDRs for allocation in the Reserved IP Ranges field. For more information, see DHCP Configuration Settings: Reference. DHCP Relay: The DHCP relay specified in the service profile will forward DHCP messages from VMs connected to the subnet to an external DHCP. Note: DHCP Relay is not supported for Distributed Transit Gateway.

4. Click **Save**.

A new VPC is created in *vCenter*.

Monitor your Virtual Private Cloud in *vSphere Client*

Learn how to monitor VPCs and information related to the current state of each VPC.

- To monitor VPCs in your *vSphere Client*, select Virtual Private Clouds, datacenter, a cluster, or a host from the inventory and navigate to **Networks > Virtual Private Clouds** folder.
- Use the drop down menu to select a **Quick Filter** or an **Advanced Filter**.
 - To use **Quick Filter**, enter value in the field.
 - To use an advanced filter, select **Advanced Filter** from the drop-down menu and click **Add New** to enter the parameters for search.

View the Topology of Virtual Private Cloud in vCenter

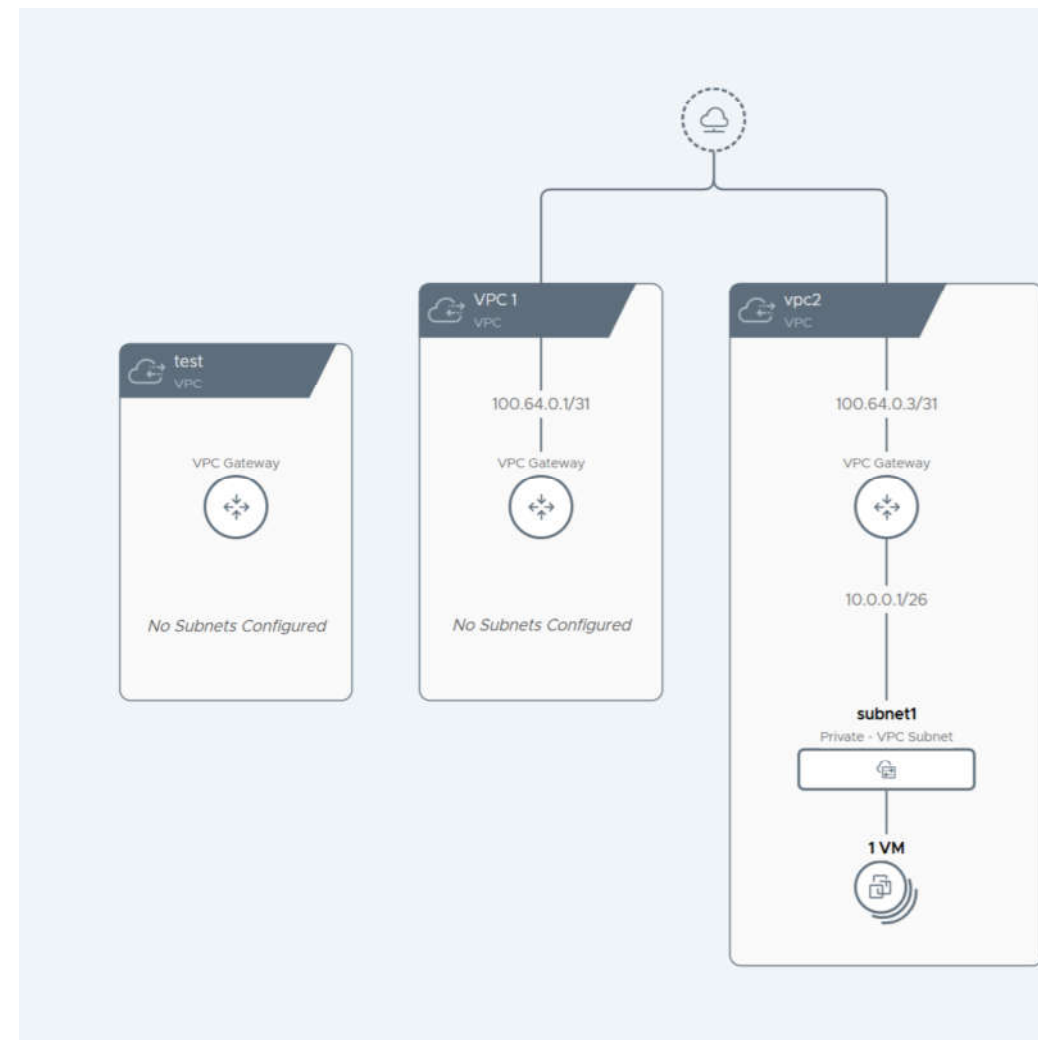
View the structure and components of a Virtual Private Cloud by using its topology diagram. The topology diagram of a VPC provides a visual representation of the VPCs and Subnets connected to the switch.

- In the *vSphere Client*, navigate to **Virtual Private Clouds** folder in your datacenter.
- Select a VPC from the VPCs are listed under **Virtual Private Clouds** folder.
- Select **Configure > Topology**.

The topology diagram of a VPC is displayed.

Figure 126: VPC Topology

Visual representation of VPCs and subnets.



View the Summary of Virtual Private Cloud in *vCenter*

Learn how to view the summary of VPCs in *vCenter*. You can view the information summary for all the VPCs or a specific VPC.

1. To view the summary of a single VPC,
 - a) From your browser, log in to the *vSphere Client*.
 - b) In the *vSphere Client*, navigate to the **VPC** under **Virtual Private Clouds**.
 - c) Click **Summary**.

Option	Description
VPC Details	
Status	Displays the status of the VPC.
VPC Subnets	Displays the number of subnets of this VPC.
External IPs	Displays the number of external IPs assigned to the VM across all datacenters and <i>vCenter</i> instances.
VMs	Displays the total number of VMs connected to this VPC across all datacenters and <i>vCenter</i> instances.
Private - VPC IP CIDRs	Block of IP CIDRs that are assigned to private subnets of this VPC. These CIDRs are local to VPCs and can overlap between the VPCs.
Connectivity and Services	
External IP Blocks	External IPs for the VMs attached to subnets. Indicates if IP blocks are set.
Private - Transit Gateway IP Blocks	Indicates if transit gateway IP blocks is set. Private IP blocks to create private subnets in the VPC.
Transit Gateway Connectivity	Default connectivity profile. Indicates if transit gateway connectivity is configured.
N-S Services	Displays the status of N-S services. Service gateway for the connected subnets and to support centralized services such as N-S firewall, NAT, or gateway QoS profile in the VPC.
Default Outbound NAT	Indicates if the traffic from workloads on private subnets are routed outside the NSX VPC.
Subnets and Workloads	
By Status	Shows the status of this VPC.
By Access Mode	Indicates the access mode for the subnet. - Private VPC - Private Transit Gateway - Public
Statistics	
North-South Traffic	Displays the statistics for all traffic in the north-south service.
Rx Dropped Packets	Total number of packets that were dropped by the receiving network.
Transmitted (Tx)	Total number of packets that were transmitted successfully.
Tx Dropped	Total number of packets that were dropped before transmission.
East-West Traffic	Displays the statistics for all traffic in the east-west service.
Rx Dropped Packets	Total number of packets that were dropped by the receiving network.
Transmitted (Tx)	Total number of packets that were transmitted successfully.
Tx Dropped	Total number of packets that were dropped before transmission.

2. To view the aggregated summary of all the VPCs,
 - a) From your browser, log in to the *vSphere Client*.
 - b) In the *vSphere Client*, navigate to **Virtual Private Clouds** in the *vCenter* inventory.
 - c) Click **Summary**.

Option	Description
Overview	
VPCs	Displays the number of VPCs
VPC Subnets	Displays the number of subnets of this VPC.
VMs	Displays the total number of VMs connected to the VPC across all datacenters and vCenter Servers.
VPCs	
Top VPCs by number of subnets	Shows the number of subnets of this VPC.

Managing your Virtual Private Cloud Subnets in *vCenter*

Learn how to manage VPC subnets in *vCenter*.

Add a VPC Subnet in *vCenter*

Learn how to add a VPC subnet in your *vCenter*.

1. From your browser, log in to the *vSphere Client*.
2. Navigate to **Virtual Private Clouds** and select the VPC.
3. From the **Actions** drop-down menu, and select **New Subnet**.
4. On the **Basic Info** page, enter the following information.

Option	Description
Name	Enter the name of the subnet.
Access Mode	Select an access mode for the subnet. - Private VPC - Private Transit Gateway - Public
Auto allocate Subnet CIDR from IP Blocks	Turn on the toggle Auto allocate Subnet CIDR from IP Blocks to auto allocate the IPs and select a subnet size. Based on whether you create a private or a public subnet, IPs are allocated from the private or public CIDRs you defined in the connectivity profile of the VPC. If you turn the toggle off, then enter the the IP CIDRs. By clicking the View IPs link, you can see the list of IP blocks with current IP addresses in use.
Subnet Size	The number of IP addresses in a network which is specified by the subnet mask.
Gateway Connectivity	Turn on this toggle to connect this subnet to the other subnets in the VPC.
DHCP Configuration - The DHCP Configuration options are available only if the DHCP settings are configured in the Service Profile linked to the VPC.	
DHCP Config	Select one of the following options:

Option	Description
	• None: DHCP will not be enabled in the subnet and all IP addresses will be set statically. • DHCP Server: The DHCP server specified in the service profile will automatically assign IP addresses and network settings to VMs connected to the subnet. You will also need to set Generic DHCP Options, and Static Bindings if you select this option. If the Auto allocate Subnet CIDR from IP Block toggle is off, then you need to provide IP CIDRs for allocation in the Reserved IP Ranges field. For more information, see DHCP Configuration Settings: Reference. • DHCP Relay: The DHCP relay specified in the service profile will forward DHCP messages from VMs connected to the subnet to an external DHCP. Note: The DHCP Relay is not supported for Distributed Transit Gateway.
External DHCP Server Addresses	This option visible only if DHCP Config is set DHCP Relay . Displays the IP addresses configured in default VPC service profile. You can edit the value of this field using NSX UI.

5. Check **Advanced Settings** to customize default DHCP configuration for the subnet.

6. For **DHCP Server**, in the **DHCP Configuration** page, enter the following.

Option	Description
DNS Server IPs	IPs of DNS servers are displayed. Read-only.
Lease Time (seconds)	The time during which clients can use IP addresses from the DHCP server. Read-only.
NTP Server IPS	The IPs of NTP servers. Read-only.
Reserved IP Ranges	Reserved IP range is used for eliminating conflicts and providing network stability.
Classic Stateless Routes	To set Classic Stateless Routes, click Add to enter the Network, then click Next, and click Add Classless Static Route. 1. Enter CIDR. 2. Enter an IPv4 address. 3. Click Add. 4. Click Apply.
Generic DHCP Options	To set generic DHCP Options, click Add to enter the code and values. Click Add Generic Option. 1. Enter Code. 2. Enter Value. 3. Click Add. 4. Click Apply.
Static Bindings	To Set Static Bindings, click Add IPV4 Static Binding to enter the following information • Name: Enter a unique display name to identify each static binding.

Option	Description
	• MAC Address: Enter the MAC address of the DHCP client to which you want to bind a static IP address. The following validations apply to MAC address in static bindings: MAC address must be unique in all the static bindings on a segment that uses a Segment DHCP server. MAC address must be unique in all the static bindings across all the segments that are connected to the gateway and which use the Gateway DHCP server. • IP Address: Required for IPv4 static binding. Enter a single IPv4 address to bind to the MAC address of the client. • Gateway Address: Enter the default gateway IP address that the DHCP for IPv4 server must provide to the DHCP client. • Host Name: Enter the host name • Lease Time: Enter the amount of time in seconds for which the IP address is bound to the DHCP client. When the lease time expires, the IP address becomes invalid and the DHCP server can assign the address to other DHCP clients on the segment. Valid range of values is 60–4294967295. Default is 86400. • DHCP Options: Set DHCP Options • Description: Add a description • Tags: Add tags to label static bindings so that you can quickly search or filter bindings, troubleshoot and trace binding-related issues, or do other tasks. 1. Click Add. 2. Click Apply.

7. Click **Next**.

8. On the warning window, read the warning and proceed.

9. For **DHCP Relay**, enter the **External DHCP Server Address**.

- a) Click **Next**.
- b) Click **Finish**.

10. If **DHCP** is set to **None**, click **Next**.

11. Click **Finish**.

Monitor your VPC Subnets in vCenter

Learn how vSphere monitors VPC subnets and provides information about the current state of each VPC subnet.

1. To monitor VPCs in your *vSphere Client*, you can
2. Navigate to inventory and navigate to **Networks > VPC Subnet**
or

3. Select a host, datacenter, cluster, VM, VPC, or a NSX Managed folder from the inventory and navigate to **Networks > VPC Subnet**.
4. Use the drop down menu to select a **Quick Filter** or an **Advanced Filter**.
5. To use the advanced filter, select **Advanced Filter** from the drop-down menu and click **Add New** to enter the parameters for search.

View the Summary of VPC Subnet in vCenter

Learn how to view the summary of VPC subnet in vCenter.

1. From your browser, log in to the *vSphere Client*.
2. Navigate to the **VPC Subnet**.
3. Click **Summary**.
4. To view the statistics, click [View More](#).
5. Cumulative statistics for all traffic through the ports on the subnet are displayed.
6. Click **VPC Gateway Interface** to view more details on data, total packets received and transmitted, and dropped packets received and transmitted.

You can view the details of the VPC Subnet.

Option	Description
Access Mode	One of the following options is present: • Private VPC • Private Transit Gateway • Public
Gateway Connectivity	It displays if this subnet is connected to the other subnets in the VPC. One of the options is present: • Yes • No
IP CIDR	Based on whether you create a private or a public subnet, IPs are allocated from the private or public CIDRs you defined in the connectivity profile of the VPC.
Gateway Address	The default gateway address.
VMs	Shows the total number virtual machines attached to this subnet across all data centers and vCenters.
DHCP Type	One of the following options is present: • None: DHCP will not be enabled in the subnet and all IP addresses will be set statically. • DHCP Server: The DHCP server specified in the service profile will automatically assign IP addresses • DHCP Relay: The DHCP relay specified in the service profile will forward DHCP messages from VMs connected to the subnet to an external DHCP.

Connect Virtual Machines to a Subnet

Learn how to connect virtual machines to a subnet. You can also connect VM to subnet by editing the VM.

1. From your browser, log in to the *vSphere Client*.
2. Navigate to **Networking > VPC > VPC Subnets**.
3. Select a VPC Subnet from your inventory.
4. From the **Actions** drop-down menu, select **Connect VMs**.
5. You can filter using a **Quick Filter** or an **Advanced Filter**.
6. Select virtual machines to connect to the subnet.
7. Select a network adapter from the **Network Adapter** drop-down menu.
8. On the **Ready to Complete** page, click **Finish**.

Assign an External IP

Learn how to assign an external IP to your virtual machines.

1. In the *vSphere Client*, navigate to VM.
2. From the **Actions** drop-down menu, select **Assign External IP**. In the **Assign External IP** page, enter the settings.

Option	Description
External IP	Click Set to set external IPs for the VMs attached to subnets. You will need to select the VM, the NIC, and whether to auto assign the IPs or provide IPs from an IP block.
Virtual Machine	Shows the virtual machines that are connected to the private subnets in the VPC.
Network Adapter	Shows the network adapters that are not assigned an external IP and connected to private subnets in the VPC.

Unassign an External IP

Learn how to unassign an external IP from your virtual machines.

1. In the *vSphere Client*, navigate to a VPC or a Subnet.
2. Click **Configure > External IPs**.
3. Select the IP listed under **External IP**.
4. Click **Unassign**.
5. Confirm the prompt by clicking **Unassign**.

SDDC Manager Workflows in VMware Cloud Foundation 9.0

With VMware Cloud Foundation 9.0 the SDDC Manager UI is being deprecated. SDDC Manager workflows can now be found in VCF Operations and vSphere Client.

- [Managing Licenses in VMware Cloud Foundation](#)
- [Managing Certificates in VMware Cloud Foundation](#)
- [Managing ESX Hosts in VMware Cloud Foundation](#)
- [Managing VCF Domains in VMware Cloud Foundation](#)
- [Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform](#)
- [Managing Network Connectivity in vCenter](#)
- [Configuring Management Appliances](#)
- [Lifecycle Management of VCF Core Components](#)

- [Lifecycle Management of VCF Management Components](#)
- [Managing Accounts and Passwords for VMware Cloud Foundation Components](#)
- [Backup and Restore of VMware Cloud Foundation](#)